Mortgage Constraints: Incidence, Adjustment, and Transmission*

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Abstract

Mortgage constraints are standard macroprudential tools, but lower credit volume says little about who bears the cost or whether financial vulnerability falls. We decompose the effect of a tightening into three components: the direct effect on households; the equilibrium incidence through house prices, rents, and credit supply; and the corrective benefit. We use this framework to synthesize household-level, housing-market, lender, cross-country, and credit-crisis evidence. The direct effect falls on young and low-wealth households, largely through fewer mortgage originations. Aggregate credit responses are several times larger than mechanically aggregated micro responses, consistent with general equilibrium amplification, although settings and outcomes differ across studies. Direct evidence on the corrective benefit is scarce, and few of the 154 tightening episodes we examine occur when credit and house-price growth is high. Estimating direct and equilibrium effects in one setting and identifying the amplification channel are the most valuable next steps.

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1 Introduction

Few macroprudential policies intervene as directly in household choices as those regulating mortgage borrowing. Limits on loan-to-value (LTV) and loan-to-income ratios, debt-service burdens, and amortization requirements ("mortgage constraints") are used in most advanced economies. Authorities justify them primarily by the need to restrain credit growth and household leverage, with financial stability as the ultimate objective. A natural evaluation of such policies therefore asks whether household credit fell after a rule was introduced. Yet a lower volume of credit is an incomplete measure of policy success for three reasons. First, it says nothing about which households gave up borrowing and what the lost borrowing capacity was worth to them. Second, it does not capture the market responses that redistribute gains and losses across buyers, owners, renters, and lenders. Third, it does not reveal whether financial vulnerability was reduced or merely relocated to other borrowers, lenders, or assets.

The evidence needed to address these shortcomings is spread across several strands of the literature. Household-level studies identify who is constrained and how those households adjust. Studies of housing markets and financial intermediaries examine changes in prices, lending conditions, and the allocation of credit. Cross-country studies estimate aggregate policy effects on credit. And a separate literature relates credit growth, house prices, and debt servicing to subsequent financial distress. These estimates are rarely interpreted together.

In this article, we provide a common framework that organizes the literature on mortgage constraints. We decompose the effect of a mortgage tightening into three components: the direct effect on households whose effective borrowing capacity falls, the incidence of the equilibrium changes that follow, and the corrective benefit from lower financial vulnerability. We use this framework to review and synthesize the literature. Along the way, we classify the objectives authorities state when introducing or adjusting mortgage constraints, compare household-level and aggregate credit estimates, and place tightening episodes on a historical map of crisis risk. We end with a discussion of where the gaps in the literature are most pressing.

Turning first to the direct effect, the household-level evidence gives a consistent picture of who is affected and how they respond. LTV limits fall mainly on households with little wealth or housing equity, while income-based limits fall on households whose desired borrowing is high relative to income (**??**). Young and first-time buyers are especially exposed under both (**??**). Some affected households still buy housing, but typically with a smaller mortgage, more equity from savings or family, or a cheaper property (**?**). On the other hand, an often sizable share of potential buyers delays the purchase or remains a renter (**??**). The same decline in borrowing can therefore reflect costly exclusion from homeownership or relatively easy substitution toward savings, family support, cheaper housing, or renting. The quantity response alone is thus a poor guide to the private burden of the policy.

The quantity response is also small relative to aggregate estimates. Applying household-level treatment effects to the exposed share of new originations and scaling by the outstanding debt stock implies a reduction in annual household credit growth of about 0.15–0.22 percentage points

for a typical tightening, mostly through fewer mortgage originations. Meta-analytic estimates corrected for selective reporting are 0.6–1.1 percentage points, three to seven times larger (**?**), and individual cross-country studies report larger effects still (**??**). Equilibrium amplification is one possible explanation for the difference. Household-level designs compare more and less exposed households within a market and therefore difference out changes in prices and credit conditions common to all households. Macro estimates retain these market-wide responses, at the cost of potentially weaker identification. The gap is consistent with amplification, although bias in the macro estimates and differences in samples and policy instruments could also contribute. It is also worth highlighting that there is substantial variation in the reforms between studies. For instance, some papers study reforms that make small adjustments to mortgage constraints, whereas other studies investigate the effect of the initial activation of policies or more blunt measures of policy change.

If equilibrium adjustment accounts for a large share of the aggregate credit response, the mechanisms that produce it determine its incidence. The evidence here is thinner than the evidence on credit, but it points to three margins. First, house-price effects of mortgage regulation are small and imprecise in cross-country studies (**?**) but larger in within-country studies (**???**). Lower house prices redistribute gains between net buyers and net sellers (**?**), suggesting that mortgage constraints have allocative effects through house prices. Lower house prices also tighten LTV constraints further through lower collateral values, which affects credit-constrained households. Second, rents appear to rise where mortgage constraints keep households out of ownership, so renters may bear part of the cost (**??**). Third, lenders reallocate credit toward less exposed borrowers, regions, and asset classes (**???**), which lowers financial vulnerability only if the new lending is safer than the lending it replaces. Few studies, however, quantify winners and losers, and the evidence on the equilibrium incidence of mortgage constraints remains largely qualitative.

The final component is the corrective benefit, which is the main rationale for mortgage constraints. Direct evidence on it is scarce. One exception is **?**, who document that UK areas more exposed to the loan-to-income limit, and therefore less leveraged, had more resilient house-price growth and fewer mortgage defaults after the Brexit referendum. Evidence on household consumption is more mixed, since lower leverage comes with smaller liquid buffers that can leave households less able to smooth consumption after job loss (**?**). At the aggregate level, the "GDP-at-risk" literature finds that macroprudential tightening improves the lower tail of the future GDP distribution (**??**), although the specific contribution of mortgage constraints is less firmly established (**?**).

A much larger literature links credit to crises and can be informative for thinking about whether the corrective effects of mortgage constraints are high or low. Long historical panels show that credit growth, and in the postwar period mortgage credit in particular, predicts financial crises (**??**), and that the risk is especially elevated when rapid household credit growth coincides with rapid house-price growth (**?**). Financial crises, in turn, are often followed by significant and persistent recessions, and in some cases deep ones (**???**). An implication of this literature is

that the corrective benefit of mortgage constraints is likely to be largest when credit and house prices grow rapidly together. We place 154 mortgage-constraint tightening episodes on a "crisis map" that sorts country-years by prior household credit growth and house-price growth. Only ten episodes fall in the region where both are growing exceptionally fast, and the Nordic reforms studied in much of the household-level literature fall in an intermediate region. This could mean that authorities acted preemptively, before credit and house prices reached the high-risk region, or that the policies were used when historical risk was low and as such had small corrective benefits. Distinguishing between these would require a counterfactual path for credit, house prices, and lending standards. The direct credit response of exposed households is thus the best-measured part of the policy effect, but it reveals neither the full private burden nor the benefit to financial stability.

We end by discussing directions for future research, organized around the three components of the framework. For the direct effect, the main gap concerns the households that are kept out of the market. Following them over time would show whether mortgage constraints delay homeownership or exclude some households permanently, as well as shed light on other adjustments that might be informative about the overall burden of the policies. For equilibrium incidence, direct and market-level responses should be estimated for the same reform, so that amplification and its channels can be observed in one setting, and studies should trace whether credit removed from the regulated segment reappears elsewhere. For the corrective effect, the external cost of household leverage, which is the main rationale for mortgage constraints, needs to be measured more directly. Following exposed and non-exposed households through downturns, for instance, would show whether lower leverage dampens the fall in spending and whether this protects the incomes of other households. Tightening episodes should also be evaluated by their effect on the subsequent path of credit and house prices, not only by the economy's position when the policy was introduced. Finally, many countries implement multiple borrower-based measures, in addition to bank-based constraints aimed at targeting the same external costs. A holistic approach to evaluating and calibrating macroprudential policies would be a useful and potentially impactful step for policy.

The article proceeds as follows. Section **??** documents the use, objectives, and design of mortgage constraints. Section **??** introduces our organizing framework. Section **??** reviews policy exposure, implementation, and household adjustments. Section **??** connects these responses to aggregate effects and equilibrium incidence. Section **??** evaluates the evidence on corrective benefits. Section **??** sets out directions for future research, and Section **??** concludes.

2 Mortgage constraints: adoption and objectives

Mortgage constraints are now a standard part of the policy toolkit, but the term covers instruments that regulate different dimensions of mortgage borrowing and have been introduced for different reasons. This section distinguishes the main policy tools, documents their spread across countries, and examines the objectives policymakers attach to them and the economic rationales

behind those objectives. We describe the statutory design of these rules only briefly. Who is exposed to them, how lenders implement them, and which households end up constrained are examined in Section **??**.

2.1 Policy tools

Borrower-based mortgage constraints regulate either how much a household may borrow at origination or how quickly the resulting debt must be repaid. The most common origination constraints link borrowing to the value of the property or to household income, either directly or through required debt service.

Loan-to-value (LTV) limits cap the mortgage relative to the value of the property, requiring the borrower to finance the remaining share with equity. For a given property value, a lower maximum LTV implies a larger down payment. Because permitted borrowing is tied to the property value, changes in house prices feed directly into borrowing capacity. A collateral constraint of this form, in the tradition of **?**, is also a common representation of borrowing frictions in quantitative housing models (e.g., **???**).

Income-based limits instead tie borrowing to household income. Loan-to-income (LTI) limits cap mortgage debt relative to income, and debt-to-income (DTI) limits cap total household debt. Unlike an LTV limit, these constraints are not mechanically relaxed for homeowners when house prices rise (**??**). Debt-service-to-income (DSTI) limits cap required debt *payments* relative to income, so their tightness also depends on interest rates, maturity, and the repayment schedule. Most mortgage defaults are triggered by cash-flow shocks rather than by negative equity alone (**?**), so DSTI limits act on a margin that is directly relevant for repayment risk.[1]

Amortization requirements and *maturity limits* operate differently: they do not cap the mortgage at origination but determine how quickly it must be repaid. They raise required payments and reduce short-run cash-flow flexibility while accelerating the accumulation of housing equity. In practice, they may also reduce borrowing at origination, because households adjust loan size to affect the repayment path relative to, for instance, their income.

Countries frequently combine several of these instruments, so the constraint that binds for a given household can change with house prices, income, and interest rates. Regulations also differ in how strictly the limits apply. Some are hard caps on individual loans, while others allow lenders to exceed the limit for a share of new lending, either through a flexibility quota (**?**) or by capping the share of high-ratio loans in a lender's new lending (**?**), as in the UK limit on mortgages at or above 4.5 times income. Many regimes also treat first-time buyers, refinancing, or new construction differently. These features determine how far the limit written in regulation translates into the limit households face, a wedge we return to in Section **??**.

[1] Terminology differs across jurisdictions and databases. In the United States, "DTI" refers to the ratio of monthly debt payments to income, which is a DSTI limit in our terminology. The rule studied by **?** is therefore a DSTI limit.

2.2 Adoption and scope

Mortgage constraints have spread widely since the early 1990s. Panel (a) of Figure **??** shows the cumulative number of countries in the Integrated Macroprudential Policy (iMaPP) database (**?**) that have used an LTV or income-based limit. Amortization requirements and maturity limits are not included. Of the 39 advanced economies in the sample, 34 (87 percent) have used an LTV or income-based limit at some point. Adoption is less universal but still substantial among emerging market and developing economies: 52 of 122 countries (43 percent) have used at least one such measure. Collateral-based limits such as LTV caps are somewhat more common than income-based limits in both groups.

Some countries, such as Hong Kong and Korea, introduced LTV or income-based limits before the global financial crisis. Adoption accelerated after 2008, however. By then, the expansion of mortgage credit supply was widely understood to have fueled the boom in house prices and household leverage and to have deepened the ensuing bust (**????**), although the relative importance of credit supply and house-price beliefs remains debated (**?**). The post-crisis wave established mortgage constraints as a central macroprudential instrument (**?**).

2.3 Objectives and rationales

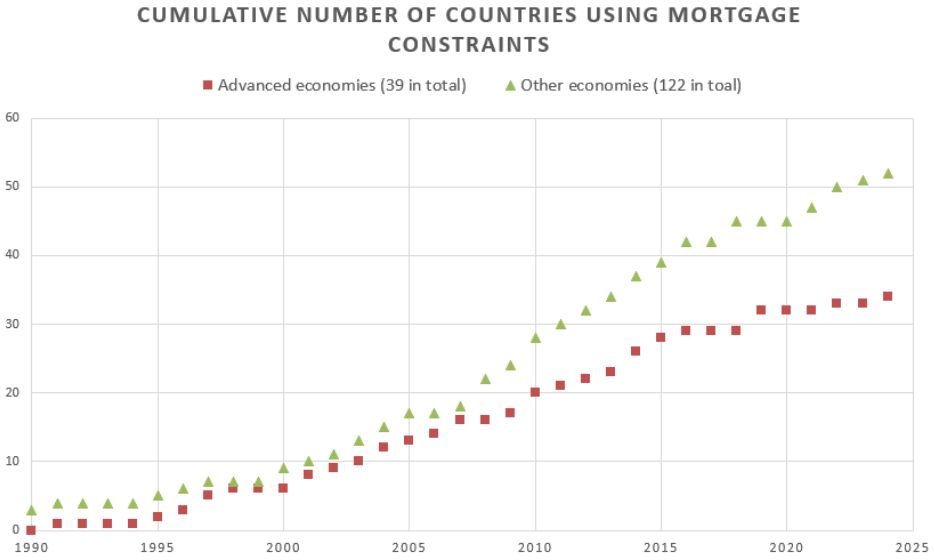
Mortgage constraints are much used, but what exactly do they set out to achieve? To systematically characterize this, we use the objectives that national authorities report in the ECB's Macroprudential Policies Evaluation Database (MaPPED), which covers 113 macroprudential actions on residential lending standards in 18 European countries. We group the reported objectives into four categories (*credit and household leverage*, *financial stability*, *bank sector*, and *house prices*) and a residual for other objectives. Authorities can report several objectives for the same measure, so the shares do not sum to 100 percent. Appendix **??** describes the data.

Panel (b) of Figure **??** shows that credit and household leverage are the most frequently cited concerns: 81 percent of measures report at least one credit-related objective, and sectoral credit growth alone is cited by 70 percent.[2] Financial-stability objectives, including resilience, appear in 42 percent of measures and house-price developments in 39 percent. Objectives referring specifically to the banking sector appear in only 8 percent. Authorities thus cite credit growth, an intermediate target, nearly twice as often as financial stability, the ultimate objective that motivates macroprudential policy.

The academic literature offers at least two broad externality-based rationales for macroprudential policy, and both appear in the motivations policymakers provide. The first is a *pecuniary externality*. Individual borrowers do not internalize that higher leverage can amplify forced asset sales following an adverse shock. The resulting decline in asset prices reduces other borrowers' collateral values and tightens their borrowing constraints (**?????**). This rationale calls for tar-

[2]Because MaPPED covers only European countries, panel (b) speaks to a narrower set of countries than panel (a).

(a) Cumulative number of countries that have used an LTV or income-based limit



(b) Stated objectives of borrower-based mortgage constraints

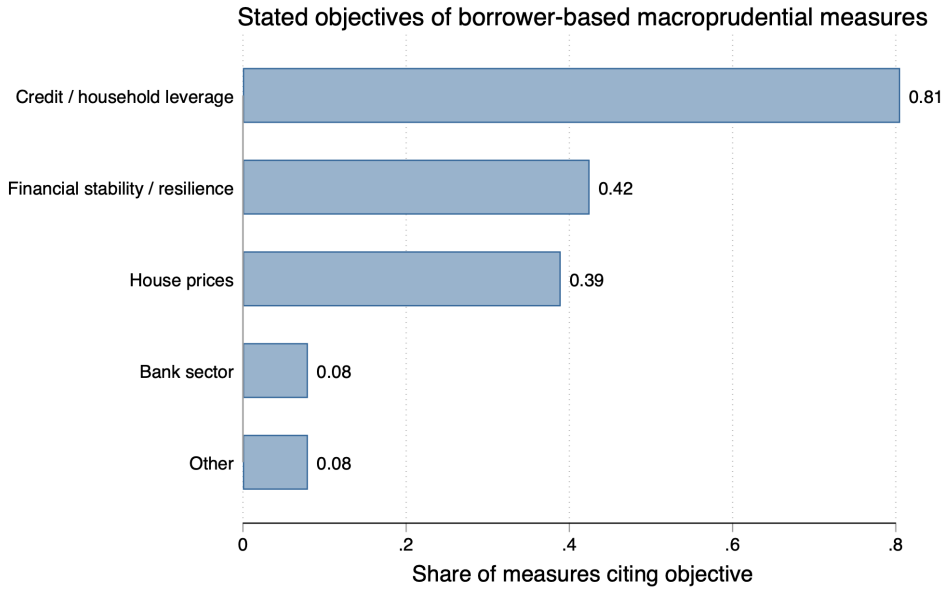


Figure 1: Adoption and stated objectives of mortgage constraints

Notes. Panel (a): data from the Integrated Macroprudential Policy (iMaPP) database (**?**). A country is counted from the first year in which it has an LTV or income-based limit in force; the sample covers 39 advanced economies and 122 emerging market and developing economies. Panel (b): share of the 113 policy actions in the MaPPED sample (**?**) citing an objective in each of the four broad categories or the residual. Objectives are multi-select and unranked, so shares do not sum to 100 percent. See Appendix **??** for the sample definition and mapping.

geting debt levels themselves, in line with, for example, the Finnish authorities' stated goal of curbing debt accumulation. The second is an *aggregate demand externality*. When monetary policy is constrained, borrowers do not internalize the effect of their borrowing decisions on aggregate demand (**?**), and deleveraging by constrained borrowers can push the economy into

a liquidity trap (**?**). Swedish and UK authorities, for instance, appeal to this rationale when they link high household debt to sharp consumption cuts after macroeconomic shocks. Both rationales correspond to what the framework in Section **??** calls the external cost of financial vulnerability. Section **??** reviews the evidence on the size of that cost.

A third rationale, consumer protection, does not rest on an externality. The US Ability-to-Repay/Qualified Mortgage rule of 2014, the rule studied by **?**, had the main rationale of strengthening mortgage underwriting practices rather than restraining credit growth or reducing systemic risk. Such a rule targets failures in private borrowing and lending decisions themselves. In our framework, this changes the interpretation of the direct effect: if households or lenders misjudge repayment risk, part of the reduction in borrowing is a private gain rather than a cost.

Given the prominence of credit and leverage among stated objectives, much of the empirical literature asks whether mortgage constraints reduce household credit growth, and our review starts there as well. But lower credit is valued because it is expected to reduce financial vulnerability, and restricting borrowing is unlikely to be costless for the households it constrains or for the markets they trade in. The next section therefore develops a framework for identifying the relevant economic margins for a more holistic overview of the effects of mortgage constraints.

3 Framework

A mortgage constraint works through a sequence of steps, summarized in Figure **??**. The statutory rule changes the effective borrowing capacity households face; households respond by adjusting their borrowing, saving, housing, tenure, or other margins.[3] These responses in turn may move house prices, rents, and lender behavior, spreading the effects beyond the households directly constrained and determining how the policy changes aggregate outcomes.

In Appendix **??**, we outline a framework to capture these adjustments. In the context of that framework, the sequence discussed above provides three types of effects that are relevant for economic welfare: the direct effect on households whose effective borrowing capacity changes, the incidence of the equilibrium changes that follow, and potentially the corrective effects on the externalities associated with less restricted or unrestricted borrowing. In this section we summarize and go through the key organizing equation that emerges from that framework, while Appendix **??** provides more details.

3.1 The organizing equation

Before stating the key organizing equation, we introduce some notation. Let τ denote the statutory policy parameter, normalized so that a higher value permits more borrowing. For an LTV constraint, τ is the maximum loan-to-value ratio. For an LTI constraint, it is the maximum loan-to-income ratio. Let $\kappa_i(\tau)$ denote household i's statutory borrowing capacity, measured in

[3] Lenders might also adjust their behavior in response to these policies, something we return to in Section **??**.

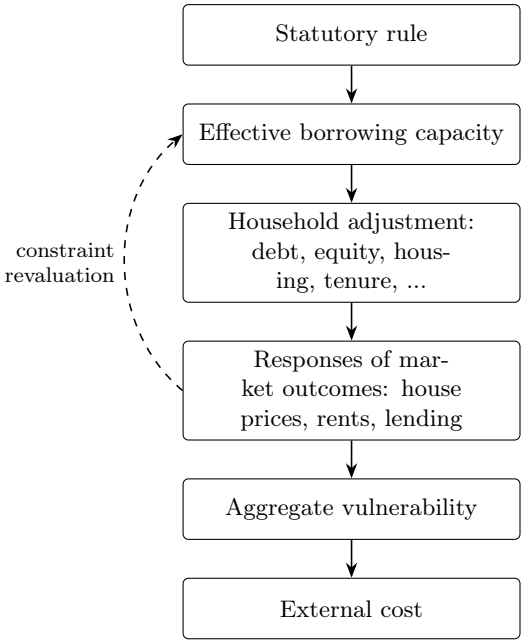


Figure 2: From the statutory rule to the three welfare terms

Notes. The figure shows the sequence of policy effects behind equation (**??**): the first three boxes correspond to term (I) (Section **??**), the fourth to term (II) (Section **??**), and the last two to term (III) (Section **??**). The dashed arrow indicates that market outcomes can feed back into effective borrowing capacity, as when lower house prices reduce collateral values under an LTV limit.

currency. Under an LTV rule, $\kappa_i(\tau) = \tau P h_i$, where P is the price per unit of housing and h_i is the amount purchased; under an LTI rule, $\kappa_i(\tau) = \tau y_i$, where y_i is the household income used by the lender when applying the rule. A change in the LTV limit thus affects statutory capacity in proportion to the property value, and a change in the LTI limit in proportion to income, so the instrument determines both the size and the distribution of statutory exposure. DSTI limits and amortization requirements fit the same notation, as Appendix **??** shows. When presenting the framework below, we will consider one generic borrowing constraint κ.

The capacity households face in practice may differ from the statutory limit. Let $\tilde{\kappa}_i$ denote effective borrowing capacity and define implementation pass-through ϵ_i by

$$\frac{\partial \tilde{\kappa}_i}{\partial \tau} = \epsilon_i \frac{\partial \kappa_i}{\partial \tau}. \tag{1}$$

Pass-through is one when the statutory change reaches the household one-for-one and smaller when exemptions, lender discretion, or other implementation margins insulate the household from it.

Let λ_i denote the shadow value of relaxing household i's borrowing constraint. It is zero when the constraint does not affect the household's attainable choices and positive when additional capacity would improve them. Because it is a marginal concept, it does not capture the loss of a household displaced from ownership altogether, a case we return to below.

Mortgage constraints can also change market outcomes. Let x_j denote an equilibrium outcome such as house prices, rents, mortgage spreads, or market-wide underwriting conditions. Let m_{ij} measure the direct money-metric exposure of household i to that outcome, holding borrowing capacity fixed. For market prices, m_{ij} corresponds to the household's net trading position. Net buyers and net sellers of housing, for example, hold opposite positions with respect to house prices, and borrowers and the owners of lending institutions hold opposite positions with respect to mortgage spreads.

Some market outcomes also feed back into borrowing capacity. This channel is captured by $\partial\tilde{\kappa}_i/\partial x_j$. Under an LTV rule, a decline in the house price reduces the collateral against which a household can borrow. Under an income-based rule, there is no corresponding mechanical effect of house prices on the statutory limit, although interest rates and spreads affect capacity under a DSTI limit.

Finally, let Z collect aggregate outcomes that potentially impose external costs. These may include aggregate household credit, the distribution of leverage and debt-service burdens, the composition of new lending, and house-price growth. Let $\Psi(Z)$ denote the external social cost of this aggregate vulnerability. It is external because individual borrowers and lenders take aggregate vulnerability as given, whereas losses they expect to bear themselves are already reflected in their choices.

Using this notation, we show in Appendix **??** that the social welfare effect of a marginal policy change can be written as

$$\mathrm{d}W = \underbrace{\left[\int \lambda_i \epsilon_i \frac{\partial \kappa_i}{\partial \tau}\,\mathrm{d}i\right]\mathrm{d}\tau}_{\text{(I) Direct effect}} + \underbrace{\sum_j \left[\int \left(m_{ij} + \lambda_i \frac{\partial \tilde{\kappa}_i}{\partial x_j}\right)\mathrm{d}i\right]\mathrm{d}x_j}_{\text{(II) Equilibrium incidence}} - \underbrace{\nabla\Psi(Z)\cdot\mathrm{d}Z}_{\text{(III) Corrective effect}} . \tag{2}$$

Equation (**??**) is a first-order welfare decomposition for a marginal policy change, derived under the assumptions that households optimize given prices and take aggregate vulnerability as given. It does not provide a welfare estimate without additional empirical and normative inputs, and the loss of households displaced from ownership altogether, which is not marginal, is discussed separately below. However, because the equation identifies the information needed to evaluate the policies from a welfare perspective, we use equation (**??**) as a starting point for selecting and synthesizing the literature. We now discuss each of the three terms in more detail.

Direct household effects. Term (I) values the policy-induced change in effective borrowing capacity. Since a tightening has $\mathrm{d}\tau < 0$ for the policy parameters considered here, the term is negative for households with $\lambda_i > 0$ whenever the statutory change passes through to effective capacity. The magnitude of this effect depends on statutory exposure $\partial\kappa_i/\partial\tau$, pass-through ϵ_i, and the shadow value λ_i.

The responses we observe in the data—using savings, buying a cheaper home, or turning to family members for help—are ways in which households cope with a loss of borrowing capacity.

They should therefore not be counted as separate costs in addition to that loss. But they are still informative. They reveal the alternatives available to constrained households, indicate how costly the constraint is, and show how the policy reshapes household balance sheets.

Households that stop buying altogether require separate treatment. Some may have been close to indifferent between owning and renting. For them, the private cost of postponing a purchase is small, even though an entire mortgage disappears from the credit statistics. Others may strongly prefer to own but are kept out by a down-payment requirement, a rejected application, or a bank that has exhausted its exemption quota. Their welfare loss can be substantial and is not captured by the marginal shadow value λ_i. We therefore treat this loss as part of the direct effect, but outside the first-order expression in equation (**??**). Credit data alone cannot distinguish between marginal households that choose not to buy and households that are effectively excluded from homeownership. Evidence on those who do not obtain a mortgage is therefore central to Section **??**.

This discussion also assumes that households make borrowing decisions in accordance with their own welfare. If borrowers instead underestimate repayment risk, a mortgage constraint may also protect them from taking on debt they later regret. This creates an additional benefit that partly offsets the direct cost of the constraint and provides a welfare rationale for the consumer-protection objective discussed in Section **??**. We develop this extension in Appendix **??**.

Equilibrium incidence. Term (II) captures welfare effects that operate through endogenous market responses, and each response $\mathrm{d}x_j$ is valued through two channels. The first is the household's market exposure m_{ij}. For instance, a decline in house prices benefits net buyers and imposes losses on net sellers, a change in rents redistributes between renters and landlords, and a change in mortgage spreads between borrowers and the owners of lending institutions. The second is the revaluation of borrowing capacity: if an equilibrium outcome enters the effective constraint, a change in it has an additional value for households with $\lambda_i > 0$. For instance, under LTV limits, a policy-induced decline in house prices reduces collateral values and may tighten capacity for existing homeowners beyond the statutory change.

The two channels aggregate differently. Every purchase is another household's sale, so for market prices the exposures m_{ij} sum to zero and the first channel is a transfer between households. Yet policymakers might attach different weights to different population groups, which is why the review documents these transfers. The second channel does not net out: a price decline that reduces collateral values lowers borrowing capacity for every constrained household, with no offsetting gain elsewhere.

Corrective effects. Term (III) values the policy-induced change in the external costs. We often refer to this as "financial vulnerability" or "financial stability." Since Z is a vector, $\nabla\Psi(Z)$ is the vector of partial derivatives of the external cost with respect to each of its components: the marginal external cost of one more unit of aggregate credit, of a higher debt-service burden, of faster house-price growth, and so on. The dot product $\nabla\Psi(Z) \cdot \mathrm{d}Z$ multiplies each of these

marginal costs by the policy-induced change in the corresponding component and sums them, which is the total change in the external cost, $\mathrm{d}\Psi$. If a tightening moves Z in a direction that lowers the external cost, then $\mathrm{d}\Psi < 0$ and the corrective effect $-\mathrm{d}\Psi$ is positive. The term thus combines two empirical relationships: the total effect of the policy on each outcome in Z, which is $\mathrm{d}Z$, and the marginal external cost of each outcome, which is $\nabla\Psi(Z)$.

4 Direct effects on households

We start by considering the first term. Statutory design does not directly determine who is constrained. Incidence also depends on how lenders implement the rule and on the resources households can draw on to meet it. We therefore first examine which households are exposed to different types of constraints and discuss how lender discretion and family support shape their effective incidence. We then review how affected households adjust their borrowing, savings, housing choices, and timing of homeownership. Together, this evidence shows who bears the direct effects in term (I) and how mortgage constraints change household credit and balance sheets.

4.1 Policy exposure

Policy exposure, $\partial\kappa_i/\partial\tau$, depends on the instrument. An LTV limit binds mainly for households with insufficient wealth or housing equity to meet the down payment, and the statutory change is largest for expensive properties. LTI and DTI limits bind for households whose desired debt is high relative to current earnings; DSTI limits additionally depend on interest rates and the amortization schedule. These differences create systematic demographic exposure. Young and first-time buyers have accumulated little wealth or housing equity and are often early in their earnings profiles. Consistent with this, they emerge across the literature as among the groups most affected by mortgage constraints (**???????**).

Exposure also changes with market conditions. Rising house prices increase the down payment needed to purchase a given home, making an LTV limit more restrictive for prospective first-time buyers. The same price growth, however, creates equity for existing homeowners and may relax their LTV constraint even when the statutory cap is unchanged. LTV incidence can therefore become increasingly concentrated outside existing homeownership during a prolonged housing boom (**?**). Interest-rate changes similarly alter the restrictiveness of DSTI limits, while income growth changes exposure to LTI and DTI limits (**?**). Consequently, incidence cannot be inferred from the headline threshold alone, as it reflects the interaction between the policy tool and the distribution of income, wealth, house prices, and existing equity.

4.2 Implementation and effective incidence

Statutory exposure does not necessarily translate into an equivalent change in effective borrowing capacity. This is captured by the pass-through parameter ϵ_i. Two observationally similar households covered by the same headline limit may face very different effective constraints depending on their lender, collateral, and access to outside resources.

Lender discretion is an important source of heterogeneity (**?**). Many mortgage regulations allow banks to exempt a fraction of originations from the statutory limits. Evidence from Ireland shows that banks manage these quotas actively rather than allocating exemptions randomly, with lower-income households in high-price areas particularly likely to receive one (**?**). The UK cap on the share of high loan-to-income originations provides particularly clear evidence of this allocation. Lenders constrained by the cap issue fewer high-LTI mortgages and charge more for them, with the largest effects on low-income borrowers. Lenders with spare capacity partly offset this contraction by expanding high-LTI lending, but primarily to other groups rather than to low-income borrowers (**?**). Discretion also extends beyond formal exemption quotas. Banks may differ in the collateral they recognize, the property valuations they accept, and the additional underwriting standards they impose. The same statutory limit can therefore translate into very different borrowing opportunities across households and lenders.

Family resources provide another important source of heterogeneity. A parental guarantee, co-borrower arrangement, or pledge of additional collateral can change the mortgage capacity recognized by the lender and thereby affect ϵ_i. A cash transfer used for the down payment operates differently, as it does not ordinarily change the statutory mortgage limit but enables the household to satisfy it with less mortgage borrowing. Both forms of support weaken the effective force of regulation from the prospective buyer's perspective.

The empirical evidence indicates that these family resources matter. **?** find that pre-death bequests became a stronger determinant of housing-market entry after Switzerland introduced LTV limits, while **?** find that wealthy parents became more predictive of entering homeownership as Norwegian LTV limits tightened. The same mechanism operates in reverse for a loosening: **?** find that the gains from the UK Help-to-Buy program were concentrated among households unlikely to have had access to family transfers. In addition, **?** show that parents borrow against their own housing to help their children enter the housing market, but that the scope varies with mortgage constraints. Tighter mortgage constraints can therefore increase the importance of family background.

4.3 How households adjust

Once effective borrowing capacity falls, households can reduce mortgage debt, contribute more equity, change the housing they consume, or delay or abandon a purchase. These are alternative responses to the same change in borrowing capacity, and they matter both for the private consequences of regulation and for how the policy changes total credit and household balance

sheets.

4.3.1 Credit adjustments

The micro literature has focused on household mortgage responses, which combine an intensive margin, that is, a smaller mortgage among households that still purchase, and an extensive margin, that is, a postponed or abandoned purchase that removes an entire origination. In this section, we discuss papers that estimate one or both margins. Based on a subset of that literature, we also aggregate their results mechanically to aggregate credit growth, holding repayments and other credit flows fixed. The purpose of the aggregation is to compare micro-implied aggregate credit effects with the direct macro effects. The details of the calculation are provided in Appendix **??**.

? study the 43 percent DTI threshold (a DSTI limit in our terminology) applying to US jumbo mortgages. Among affected high-DTI borrowers, 20 percent shift to DTI ratios below the threshold and 15 percent leave the market. For the average borrower above the threshold, reaching 43 percent would have required a loan roughly 4.4 percent smaller, which the authors argue is too large to reflect borrower demand and instead points to lenders withdrawing supply. Because the rule applied only to the jumbo segment, its realized effect on aggregate credit is negligible. **?** study Norwegian LTV limits using predicted pre-reform exposure. The probability of purchasing falls by 6 percent among affected households, while debt among continuing buyers falls by approximately 11 percent. **?** combine bunching and predicted-treatment designs for Finland's LTV limit on first-time buyers. Conditional on purchasing, affected households reduce mortgage size by about €6,500, or roughly 5 percent, but the decline in entry is quantitatively more important. Effects similar to those in the papers discussed above are also found in other countries and settings (**??**).

Back-of-the-envelope aggregations of the estimates discussed above yield a drop in aggregate lending growth of about 0.15–0.22 percentage points when mortgage constraints tighten. These effects appear modest in aggregate. In the US and Finnish cases, most of the effect comes from fewer mortgage originations; in Norway, where 6 percent of treated households do not purchase and continuing buyers borrow 11 percent less, the two margins are of similar size. Moreover, all of these estimates are accounting exercises based on within-market comparisons. Only **?** report an aggregate calculation of their own, and **?** extrapolate from the jumbo segment to the full purchase market, so the common-scale numbers we use in Section **??** rest on our own mapping (Appendix **??**) and are approximate.

Amortization requirements can also reduce borrowing at origination even though they regulate the repayment path. Exploiting a Swedish reform that eliminated interest-only mortgages above a loan-to-value ratio of 50 percent, **?** document substantial bunching below the threshold and approximately 5 percent lower borrowing, concentrated among relatively wealthy households with substantial unused borrowing capacity, which points to sensitivity to required monthly payments rather than to a binding constraint.

4.3.2 Other adjustments

Changes in mortgage debt provide only a partial account of how households respond to borrowing constraints. Households may also rebalance their financial portfolios, alter consumption, change the housing bundle they purchase, delay or abandon homeownership, and adjust their behavior following adverse income shocks. These responses are important in their own right, but they are also informative about the overall burden of the policies, as well as how the policies affect financial fragility.

Portfolio rebalancing. LTV constraints can lead households to replace mortgage debt with housing equity financed from their own assets. Both **?** and **?** find that LTV-constrained buyers draw down liquid assets when purchasing a home. **?** show evidence suggesting that such a drain of liquidity can be costly to households. The decline is temporary among Dutch buyers, whereas Norwegian buyers remain less liquid for at least three years. In principle, households could also finance the additional equity contribution by selling less liquid financial assets. However, **?** find no corresponding response in other financial assets, suggesting that the adjustment in their setting is concentrated in readily available liquidity rather than reflecting a broad reallocation of the financial portfolio. Whether the loss of liquidity leaves these households less able to absorb later shocks is part of the corrective effect, and we return to it in Section **??**.

Consumption in normal times. Households could respond to tighter borrowing constraints by lowering consumption. However, this seems not to be the case. **?** find no discernible effect on consumption among affected Norwegian buyers in normal times. Importantly, their results apply to households that purchase a home despite the constraint. They do not capture consumption effects among households that delay or abandon a purchase. Such households may save more to reach the new down payment requirement, or they may save less if purchasing in the near future becomes unattainable. **?** find evidence of the latter channel in a quantitative model studying LTV caps.

Housing quality and location. Households can also respond by changing the housing bundle they purchase. This margin includes not only dwelling size and quality, but also location, commuting time, neighborhood characteristics, and access to employment and amenities. It is the primary response of LTI-constrained households in **?**, and it is present under LTV limits as well. Other studies, such as **??**, find evidence pointing toward households buying cheaper houses. Using Israeli data, **?** shows that households affected by an LTV limit purchase cheaper and smaller homes and move farther from central Tel Aviv. These responses demonstrate that mortgage constraints can alter real housing consumption even when a household remains a homeowner. A reduction in purchase price should therefore not automatically be interpreted as a proportionate welfare loss: it may reflect lower physical quality, a less preferred location, or simply a relatively inexpensive substitute. Nevertheless, changes in commuting distance, amenities, and neighborhood socioeconomic composition indicate that the consequences may

extend beyond the financial characteristics of the mortgage.

Transition to homeownership. For some households, adjustment occurs not through the characteristics of the mortgage or property, but through whether and when they purchase a home. This is the extensive margin that dominates the credit response in Section **??**, viewed from the household's side. **?** and **?** document declines in purchases among affected households. **?** find that the Finnish LTV limit reduced first-time entry by 16 percent among affected households, or about 8 percent of all first-time apartment buyers with a mortgage, with the response concentrated among households below median income. **?** similarly document reduced entry following Swiss LTV regulation. The extensive housing response is therefore quantitatively important and unevenly distributed: young, low-income, and first-time buyers are particularly likely to experience delayed access to ownership.

The welfare interpretation of these effects depends on the available alternatives. If the marginal household is close to indifferent between buying now, buying later, and renting, a substantial entry response can coexist with a relatively small private cost per household. Conversely, down-payment indivisibilities, credit denial, long delays, imperfect rental substitutes, or a strong preference for ownership may make the cost substantial. **?** discuss the benefits of homeownership along several channels, pointing to potentially large costs for those who delay or forgo becoming homeowners.

Labor-market search. Lower leverage may also improve labor-market resilience. **?** show that job seekers with recourse mortgages in depressed housing markets apply for fewer jobs that require relocation.[4] Consistent with this, **?** find that displaced homeowners affected by the Norwegian LTV limit search longer and more broadly, switch occupations and industries more often, and obtain better-paid jobs. These results suggest that lower leverage reduces the pressure to accept a new job quickly, although this evidence again does not capture effects operating through delayed or forgone homeownership.

The policy tool can influence the margin of adjustment. The literature also highlights that the policy tool matters for the margin along which households adjust. **?** shows that Irish households respond differently to LTV and LTI limits. LTV-constrained households contribute more equity and largely preserve the value of the property purchased, whereas LTI-constrained households primarily purchase cheaper properties. The form of the constraint therefore determines whether adjustment occurs mainly through household balance sheets or through housing demand. This result frames the evidence above: the liquidity responses discussed earlier are largely an LTV phenomenon, while the housing-bundle responses are more prominent under income-based limits.

[4] See also **?**.

4.4 What the evidence establishes about term (I)

Taken together, these findings can be summarized in three main sets of results. First, the direct incidence is concentrated rather than widespread. Young, first-time, and low-wealth households are most exposed, and those without family resources have fewer ways to mitigate a tightening. Second, adjustment occurs along real and financial margins that reveal how much the constraint binds: households deplete liquidity, alter housing quality and location, and delay or forgo ownership. Third, policy design matters. LTV and income-based limits expose different households and induce different substitutions, while lender implementation determines who ultimately faces the constraint.

The observed adjustments indicate that additional borrowing capacity is valuable to a non-negligible group and suggest that its value varies with liquid wealth, family support, and available housing alternatives. Quantifying the overall welfare effect is challenging, but a few studies make progress. **?** elicit households' willingness to pay for a comparable home under different down-payment requirements and other financing conditions. **?** use the introduction of an LTV cap in Sweden to quantify the willingness to pay for unregulated debt that can be used to bypass the LTV cap, finding that homebuyers are willing to pay 7.3 SEK to avoid 1 SEK of equity down payment. These studies show how financing constraints affect housing demand, but they do not identify willingness to pay for an additional unit of effective borrowing capacity for all constrained borrowers. In principle, that would require offering constrained households small increases in capacity at randomly varying fees. Mortgage-pricing notches provide a related revealed-preference design (**??**), but translating mortgage-demand elasticities into the shadow value of a regulatory constraint requires a model, and such designs generally cover only households that obtain a mortgage. A natural next step is therefore a common money-metric valuation that also includes households who delay or forgo a purchase.

5 Aggregate effects and equilibrium incidence

The household-level evidence in Section **??** compares households with different exposure to mortgage constraints while holding common market conditions fixed. A policy tightening can, however, change those market conditions. Lower housing demand may affect house prices and rents, while lenders may adjust mortgage pricing, underwriting, and the allocation of credit. These responses spread the effects of the policy beyond the households whose borrowing capacity changes directly.

In term (II) in the framework, market adjustment matters in two ways. First, changes in prices and lending conditions create gains and losses according to the positions m_{ij} that households hold. A fall in house prices, for example, benefits prospective buyers but reduces the wealth of existing owners, while higher rents redistribute from tenants to landlords. Second, a market response can feed back into effective borrowing capacity through $\partial\widetilde{\kappa}_i/\partial x_j$. Section **??** reviews these market responses and what we know of their incidence.

The same adjustments determine how the direct credit response documented in Section ?? translates into the total change in household credit. Section ?? therefore turns to aggregate credit growth, the outcome most closely aligned with policymakers' stated objectives and the key stability-relevant variable entering term (III). Comparing aggregate estimates with the direct effects implied by household-level studies reveals a sizable aggregation gap, which is consistent with quantitatively important general equilibrium amplification but does not identify it.

5.1 Market adjustment and equilibrium incidence

We organize the evidence around three market outcomes: house prices, rents, and mortgage terms and credit allocation.[5] For each, the relevant questions are whether the policy changes the market outcome, who gains and loses from that change, and whether it feeds back into borrowing capacity or aggregate credit.

5.1.1 House prices

Mortgage constraints can lower house prices by reducing how much prospective buyers are able to bid, and whether they bid at all. The impact on prices, however, depends critically on market segmentation and housing supply elasticities. This is evident in theoretical models, where the impact of credit access on house prices varies strongly with modeling assumptions. In **?**, for instance, a change in credit conditions moves homeownership and leverage but has little effect on prices, because the option to rent leaves few households constrained in their consumption of housing services. In **?**, by contrast, credit conditions affect the housing risk premium and have much larger price effects. **?** relate these differences in results to the degree of segmentation between owner-occupied and rental housing. With a frictionless rental sector, as in **?**, landlords absorb the demand of constrained buyers and prices are pinned down by rents, whereas with segmented markets credit-driven demand moves prices.[6] **?** offers a longer discussion of how credit affects house prices in structural models.

The empirical evidence on house-price effects is mixed. **?** survey a large number of studies and find weaker and more imprecise effects of macroprudential policy on house prices than on credit. Cross-country LTV tightenings reduce household credit by about 6 percent and real house prices by about 8 percent in **?**. Within-country studies based on micro data generally find larger, negative effects. They use cross-sectional variation for sharper identification, but do not necessarily capture aggregate effects. See **?** and **?** for surveys. **?** show that the Irish LTV and LTI limits slowed annual house-price growth in the urban markets most exposed to the rules

[5] The interaction with monetary policy is another possible aggregate channel. A mortgage tightening that weakens housing demand and inflationary pressure may induce a more accommodative policy rate, while lower household leverage may change the sensitivity of consumption to future rate movements. **?** study the effect of changes in maximum LTV ratios on output and inflation at the aggregate level, finding that there is little impact of changes in numerical maximum LTV ratios in advanced economies. The effect is instead concentrated in emerging countries.

[6] **?** report that their result is robust to alternative degrees of segmentation and attribute it instead to the option to rent and to long-term mortgage contracts.

from well above 20 percent to around 4 percent, while growth in less exposed markets remained stable. Their results further point toward a role for expectations, as Irish households revised their future house-price expectations downward in response to the policy. **?** also find lower Irish prices through the credit-availability channel. **?** study a mortgage constraint *loosening* and estimate that the introduction of interest-only mortgages in Denmark raised house prices by about 36 percent over the subsequent boom.

Several studies exploit regional variation in mortgage constraints or policies, or variation in the treatment of different submarkets. In New Zealand, **?** estimate relative price effects of around 3 percent by comparing regulated properties with exempt new construction, whereas **?** finds no significant response in the national house-price index. **?** show that a targeted increase in down-payment requirements reshaped asking prices and bidding around the regulatory threshold in Canada, but find no differential decline in house-price growth across more- and less-exposed Toronto neighborhoods.

? use an LTV tightening that only applied to selected districts of Taiwan to compare sales on either side of district borders. After the 2014 tightening, prices on the treated side fell by about 6 percent relative to the untreated side and did not recover when the limits were lifted. Their estimates imply a leverage–house price elasticity between 0.75 and 1. In Hong Kong, more expensive properties faced lower LTV caps, and **?** uses this variation to estimate an elasticity of house prices with respect to the LTV limit of about 0.8. In Israel, higher capital charges on high-LTV mortgages lowered the prices of homes likely to be bought with such loans by about 2.5 percent, with larger effects in expensive regions (**?**).

In sum, theory tells us that the house-price responses should be largest where housing supply is inelastic, owner-occupied and rental markets are segmented, and constrained buyers represent a large share of demand. While the empirical evidence on house-price effects is somewhat mixed, it seems to support a qualified conclusion. Mortgage constraints do not always lower national house-price growth, but studies tend to find that they can produce sizable price responses in exposed markets, roughly in proportion to the tightening in leverage where limits target specific locations or price segments.

The benefits of lower house prices are not evenly distributed. Prospective buyers benefit, while existing owners and net sellers experience capital losses (**?**). Because a homeowner who stays in place is hedged against changes in the cost of housing (**?**), a price change is mainly a transfer between those who will buy and those who will sell rather than a change in aggregate wealth (**?**). Younger and lower-wealth households are more likely to be prospective buyers, whereas older and wealthier households own a larger share of the housing stock. A decline in prices can therefore offset part of the direct burden on prospective buyers. In terms of feedback effects, lower house prices directly reduce mortgage capacity under an LTV limit by lowering collateral values for existing owners who buy again (**??**). This collateral feedback is absent from a pure LTI or DTI limit, although house-price changes may still affect desired borrowing and the distribution of households across constraints.

5.1.2 Rents

Mortgage constraints may also affect rental markets. Households that delay or abandon a purchase add to rental demand. At the same time, investors may purchase some of the properties no longer bought by prospective owner-occupiers and make them available for rent. The effect on rents depends on which response dominates and on the elasticity of rental supply. Rules that also effectively restrict mortgages for secondary homes or investment properties may place additional upward pressure on rents by limiting the supply response.

The empirical evidence is limited but consistent. For mortgage constraints themselves, direct evidence comes from the Irish setting. **?** apply the county-level exposure measure of **?** to rents and find that, in the counties where the 2015 limits were most binding, house prices grew less and rents grew more over the following two years: a one-standard-deviation increase in exposure is associated with 4.2 percent lower house-price growth and 2.5 percent higher rent growth. The authors calibrate a life-cycle model to the Irish market in which the limits raise rents by about 4 percent at the peak and 3 percent in the long run, with the losses concentrated among young and middle-income households. Evidence on broader credit tightenings points the same way. In the United States, metropolitan areas whose lenders tightened mortgage standards more after 2008 saw homeownership fall and rents rise, with annual rent growth about 2 percentage points higher than it would otherwise have been (**?**).

The estimated increase in rents also speaks to the segmentation question raised in Section **??**. Rents rising while prices fall in the exposed Irish counties indicates that rental supply did not expand in step with rental demand, the same frictional rental market that **?** infer from the response of price–rent ratios to credit shocks in the United States. The Irish result thus illustrates how a policy that lowers the price of owner-occupied housing can nevertheless leave prospective buyers worse off, given that they remain in the rental market for some time.

Higher rents transfer resources from tenants to landlords and may therefore reinforce the distributional pattern of the direct effects. The households most likely to remain renters because of a mortgage constraint are young and have relatively little wealth, while rental property is disproportionately owned by wealthier households. Rent changes do not mechanically alter mortgage capacity under an LTV or LTI limit, but they can do so indirectly by affecting disposable income and the rate at which prospective buyers accumulate a down payment. Despite the importance of this channel, the reduced-form evidence on rents remains limited and few studies consider the supply side of rental markets.

5.1.3 Mortgage terms and credit allocation

Lenders can respond to mortgage constraints by changing interest rates, fees, underwriting standards, and the allocation of lending across borrowers, regions, and asset classes. The UK cap on the share of high-LTI originations, discussed in Section **??**, shows that the policy changes both the price of mortgage credit and who receives it. Constrained lenders charged more for

high-LTI loans, and lenders with spare capacity expanded such lending mainly to borrowers other than low-income ones (**?**).

The Irish evidence shows that reallocation can extend beyond the regulated segment. Banks most exposed to the LTV and LTI limits shifted mortgage lending from lower-income to higher-income households and from urban to rural markets. They also increased lending to riskier corporate borrowers and expanded their holdings of high-yield securities (**?**). Reallocation also crossed borders: Irish banking groups took on more leveraged mortgage lending in the United Kingdom after the Irish limits were introduced, with the largest increases in LTV and LTI ratios for first-time buyers (**?**). The initial reduction in constrained mortgage lending was therefore partly accompanied by additional credit elsewhere in bank portfolios.

These responses spread the incidence of the policy. Borrowers who face higher mortgage rates or lose access to credit are worse off, while households and firms that receive reallocated lending may benefit. Bank owners may gain or lose depending on how the policy affects margins, loan volumes, and portfolio risk. Reallocation also matters for the corrective benefit. A reduction in high-LTV mortgages lowers financial vulnerability only to the extent that the displaced credit does not reappear in an equally or more fragile form.

5.2 From direct to aggregate credit effects

The preceding subsection identified several channels through which the direct credit response may be amplified. Due to the intimate connection with the need for borrowing, market-level house-price adjustment is a leading candidate. By reducing the purchasing capacity of constrained households, a tightening can put downward pressure on house prices. Under an LTV limit, lower prices also create a feedback effect on mortgage capacity. Lender responses may reinforce or offset this mechanism. We now turn to macro estimates of the effect of mortgage constraints on *credit*. Comparing these estimates with the household-level effects in Section **??** reveals a sizable aggregation gap and illustrates the importance of market-level adjustments for understanding the overall effect on credit, which is a key input into the overall effectiveness of mortgage constraints for reducing external costs.

5.2.1 Macro estimates of credit effects

The macro literature estimates the relationship between mortgage constraints and aggregate credit growth using variation across countries and over time. This avoids extrapolating from a particular group of exposed households, but it creates other comparability and identification problems. Some studies compare periods in which a constraint is in place with periods in which it is absent, while others study discrete tightening and easing actions. Policy measures, credit aggregates, and estimation horizons also differ.

? use indicators for whether borrower-based constraints are in place. In advanced economies, annual real household credit growth is 1.4 percentage points lower when a strictly enforced LTV cap is in place and 1.8 percentage points lower when a DTI limit is in place, although the LTV estimate is imprecise. The point estimates for emerging economies are several times larger. **?** instead record tightening and easing actions. They estimate that an additional LTV tightening lowers annualized housing-credit growth in the following quarter by about 1.4 percentage points, and a DSTI tightening by about 2 percentage points.

Meta-analyses produce smaller estimates. Combining results across 58 studies, **?** find that a typical macroprudential tightening, which includes a number of tools outside of mortgage constraints, lowers year-on-year total credit growth by roughly 0.6 percentage points. **?** focus on borrower-based measures and convert the estimates to a common outcome. Their unadjusted means imply reductions in annual credit growth of about 0.9 percentage points for LTV measures, 1.2 percentage points for income-based measures, and 2 percentage points when the two are used jointly. After adjusting for selective reporting of estimates with the expected sign and statistical significance, the pooled range across instruments is approximately 0.6–1.1 percentage points. The macro literature therefore consistently points toward lower credit growth after mortgage constraints are tightened. We use 0.6–1.1 percentage points as the preferred macro range because it places the estimates in the literature on a common scale and adjusts for selective reporting, while retaining the larger estimates from individual cross-country studies as an indication of the broader uncertainty.

5.2.2 The aggregation gap

The aggregate estimates are substantially larger than the direct effects documented in Section **??**, which implied reductions in annual aggregate credit growth of approximately 0.15–0.22 percentage points. These calculations include both smaller mortgages among continuing buyers and the loss of entire originations when households do not purchase. The preferred macro estimates are about three to seven times larger, and the upper end of the broader macro literature implies an even wider gap. We refer to this below as the "aggregation gap."

The aggregation gap is relevant for both remaining terms of the framework. If market adjustment accounts for a meaningful share of the larger aggregate response, the household-level evidence captures only part of the policy's incidence, making term (II) quantitatively important. The total credit response also enters term (III), where it is valued according to its effect on financial vulnerability. Understanding the source of the gap therefore matters for both who bears the effects of the policy and how large its potential corrective benefit may be.

The gap should not, however, be automatically interpreted as an estimate of general equilibrium amplification. The micro and macro studies differ in their samples, countries, policy changes, credit measures, estimation methods, and time horizons. Moreover, macro estimates may also capture anticipation effects, concurrent policy changes, or other developments around the time of a tightening, while naive extrapolation of micro results to the aggregate economy is a rough

approximation at best. The cross-sectional comparisons in micro studies might also suffer from contaminated control groups (see, for instance, **?**). With these caveats in mind, we discuss what share of the aggregation gap could plausibly result from general equilibrium amplification mechanisms such as house-price declines in the next subsection.

5.2.3 How much amplification would close the gap?

Direct estimates of equilibrium amplification arising from changes in mortgage constraints are rare. It is therefore instructive to explore general equilibrium amplification to other credit shocks and see whether these shocks generate amplification on the order of what is implied by the estimates discussed above. We express each estimate below as the ratio of the total response to the direct response, where a ratio of one indicates no amplification.

? estimate that the total regional response to the US credit expansion of the 2000s was approximately 4.5 times the direct response identified within local markets. **?** studies a lending cut at a large German bank and finds that the total county employment response is approximately 2.6 to 3.9 times the direct effect on borrowing firms, so local general equilibrium effects account for roughly 60 to 75 percent of the total. Using a different approach, **?** separates propagation through house prices from propagation through lender balance sheets for US counties during the same credit expansion.[7] His estimates imply a ratio of total to direct effects of about 5.6, ranging from 4.2 where housing supply is elastic to 7.7 where it is inelastic.[8]

The estimates above are regional in nature, and regional effects need not coincide with national effects. They may overstate aggregate amplification when housing demand or lending shifts from exposed to less exposed areas, but understate it when time fixed effects absorb common national changes in house prices or bank funding. Geographic openness matters as well. Credit shocks spill across regions when households purchase housing elsewhere, making national effects larger than local estimates, while international leakage may weaken this amplification in small open economies. Consistent with this, **?** find larger credit multipliers in less open metropolitan areas, and **?** find that aggregate house-price shocks have larger employment effects than local shocks. Monetary policy can provide an offset: euro-area mortgage regulations are unlikely to move the common policy rate, whereas central banks in Norway or Sweden may ease in response to weaker housing demand (**?**).

[7] **?** reports the multiplier as the general equilibrium response net of the direct effect: 2.90 on average, and 1.99 and 4.22 in counties with elastic and inelastic housing supply, respectively. This uses Sarto's more conservative decomposition. To make the numbers comparable to those of **?** and **?**, we add Sarto's estimated direct effect of 0.631 from Table 2 to the extra effect $\hat{M}$ in Table 7 and divide by the direct effect. The ratio of the total to direct effect is then $(2.9 + 0.631)/0.631 = 5.6$.

[8] While informative, these estimates cannot be directly used to provide a multiplier for mortgage constraints. They concern different shocks and outcomes, and most measure amplification within regions rather than at the national level. The magnitude is also unlikely to be constant across time and space. The evidence in Section **??** suggests that housing-market feedback is stronger when supply is inelastic, owner-occupied and rental markets are segmented, and constrained buyers make up a large share of demand. Intermediary amplification is likely to be stronger when mortgage lending is concentrated among a small number of highly exposed banks. Initial conditions matter as well. A tightening introduced during a housing boom, when leverage is high and many households are close to the regulatory limit, is likely to generate a larger total response than the same reform in a less stretched market.

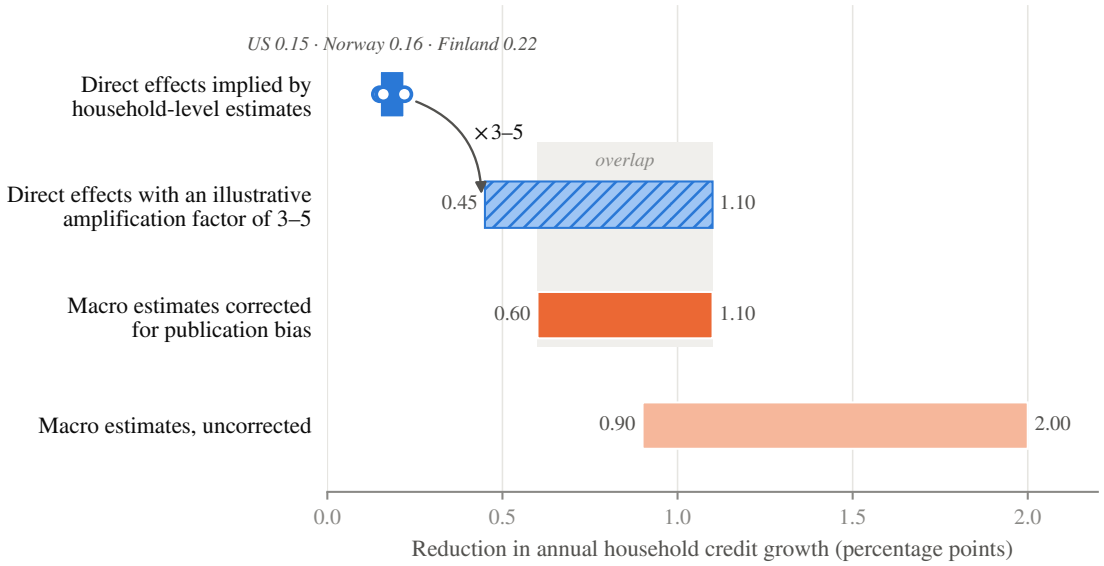


Figure 3: Illustrative magnitudes required to reconcile micro and macro estimates

Notes. The household-level estimates are back-of-the-envelope aggregate implications of **?** for the United States, **?** for Norway, and **?** for Finland. The amplified range applies factors of three to five to these estimates. The macro ranges summarize the studies reviewed in Section **??**; the narrower range is the publication-bias-adjusted estimate of **?**. The comparison is illustrative because the underlying studies differ in countries, policy tools, periods, outcomes, and research designs.

Lender networks can also move the regional-to-national correction in either direction. A shock that reduces lender net worth may contract credit across all markets served by the affected banks, causing regional estimates to understate aggregate amplification. **?**, for example, finds substantial amplification from shocks originating in other regions. By contrast, a borrower-based limit that removes lending opportunities without reducing bank capital may induce lenders to redeploy credit toward less exposed regions, borrowers, or asset classes. Such reallocation makes regional effects larger than the aggregate contraction, as illustrated by Ireland, where mortgage issuance changed little after the 2015 limits (**?**). The correction from regional to national effects therefore depends on whether the dominant mechanism is propagation or reallocation, and mortgage constraints may activate both simultaneously.

With these caveats noted, Figure **??** uses the estimates from related credit settings to illustrate how amplification could affect the comparison. Applying multipliers of roughly three to five to the household-level effects of 0.15–0.22 percentage points produces total effects of 0.45–1.1 percentage points. This range closely overlaps the preferred macro estimates of 0.6–1.1 percentage points, although it does not reach the largest estimates from individual cross-country studies. Because the benchmarks come from other types of credit shocks, the exercise serves as a plausibility check rather than an estimate of amplification from mortgage constraints. Its more limited conclusion is that the equilibrium effects required to reconcile the household-level estimates with the preferred macro range are not dramatically larger than those found elsewhere in the credit literature, but, importantly, they are yet to be estimated.

5.3 What the evidence establishes about term (II)

The evidence suggests that the market responses collected in x in term (II) may be quantitatively important, but also clearly illustrates that more evidence is necessary on the size of spillovers within a single market. Aggregate credit estimates are several times larger than the effects obtained by extrapolating from household-level studies. This gap does not directly identify general equilibrium amplification, as the underlying studies differ in their settings, methods, and outcomes. It is nevertheless consistent with sizable responses in house prices, lending conditions, and credit allocation. Evidence from related credit shocks also shows that general equilibrium effects of this magnitude are plausible.

The incidence of these responses depends on the channel. Lower house-price growth benefits prospective buyers, who tend to be younger and less wealthy than existing owners. This may offset some of the adverse distributional effects identified in term (I). House-price declines also reduce housing wealth for existing owners and, under an LTV limit, can lower the collateral available to homeowners who wish to move or refinance. This latter effect enters term (II) through the term $\partial\widetilde{\kappa}_i/\partial x_j$, and is absent from a pure income-based limit. A fall in house prices can therefore benefit first-time buyers while tightening borrowing capacity for existing owners.

Other market responses may, however, reinforce the direct burden. Higher rents fall on households that remain in the rental market because they cannot buy, potentially offsetting part of the gain from lower house prices for prospective buyers. The distributional incidence of lender reallocation is less clear. Credit that is withdrawn from highly leveraged borrowers or expensive regions may be redirected toward households with lower leverage, borrowers in cheaper areas, or firms and assets outside the regulated market. These groups are generally less exposed to the direct effect in term (I). The available evidence therefore shows that term (II) can redistribute the effects of mortgage constraints across households and markets in ways that differ from direct statutory exposure. What it does not yet provide is a joint estimate of these gains and losses, or a money-metric estimate of term (II) as a whole.

6 Corrective effects

We now turn to the final term in our organizing equation. Term (III) captures the macro-financial rationale for mortgage constraints: a tightening generates a corrective benefit when it moves the stability-relevant outcomes in Z, such as credit growth, the composition and servicing burden of debt, and house-price growth, in a direction that lowers the probability or severity of financial distress. In principle, the effect could also be negative, for instance if mortgage constraints deplete households' liquid buffers. In this section, we survey the literature on the corrective effects of mortgage constraints. We separate the discussion into three parts. First, we discuss which outcomes matter for financial stability, focusing on the role of credit in particular, in line with the literature. This captures the term $\Psi(Z)$ in term (III). Second, we consider direct changes in stability-relevant outcomes Z, including—but not limited to—household credit.

Finally, we discuss reduced-form evidence of the full term (III), coming from the GDP-at-risk literature.

6.1 What matters for the corrective benefits?

A large literature focuses on the ex ante prediction of financial crises. A central insight from that literature is that credit expansions are one of the most robust predictors of financial crises in long historical panels (**?**). In the postwar period, mortgage lending accounts for much of this relationship (**?**). Increases in household debt relative to GDP also predict weaker subsequent output growth across countries (**?**).

Panel (a) of Figure **??** illustrates, however, that this relationship is state dependent, using estimates from **?**. The cells report the probability that a financial crisis begins within three years for country-years sorted by prior household credit growth and real house-price growth. Crisis probability is 36.8 percent when the three-year change in household debt relative to GDP is in its top quintile and real house-price growth is in its top tercile, the region **?** call the R-zone ("red zone"), against 3.3 percent in the middle cell and roughly 7 percent in country-years where neither credit nor house prices are elevated. In their regression decomposition, unusually high credit growth alone adds about 9 percentage points to the three-year crisis probability and unusually high house-price growth alone adds essentially nothing, but when both are high the probability rises by about 30 percentage points. Risk is therefore particularly high during joint credit and house-price booms.

Other evidence reinforces this interpretation. The housing booms that end most badly are leveraged booms rather than equity-price booms or housing booms without credit growth (**?**), and booms financed at compressed spreads are followed by particularly severe outcomes (**?**). Credit growth under stable lending standards contains much less information (**?**). Not only does the overall credit volume matter, but so does its composition. Credit expansions concentrated in household and non-tradable-sector lending are followed by growth reversals in a way that lending to tradable firms is not (**?**).

Credit also matters for the macroeconomy once a crisis has occurred. Financial-crisis recessions produce much deeper and more persistent output losses than normal recessions, particularly after credit-intensive expansions (**?**), and in a broad panel of countries output remains around 7.5 percent below its prior path a decade after a banking crisis (**?**). Credit growth deepens recessions even when no crisis follows: a 10-percentage-point larger three-year rise in credit to GDP before a recession is associated with about 2 percent lower output three years after the peak, with household credit mattering at least as much as corporate credit (**?**). **?** temper this picture for advanced economies. Using a narrative measure of financial distress for 24 OECD countries from 1967 to 2012, they find that output falls by about 6 percent at the trough and remains roughly 5 percent below its prior path after five years, a persistent but moderate loss.

The predictive content of credit and the depth of financial crises suggest that curbing credit

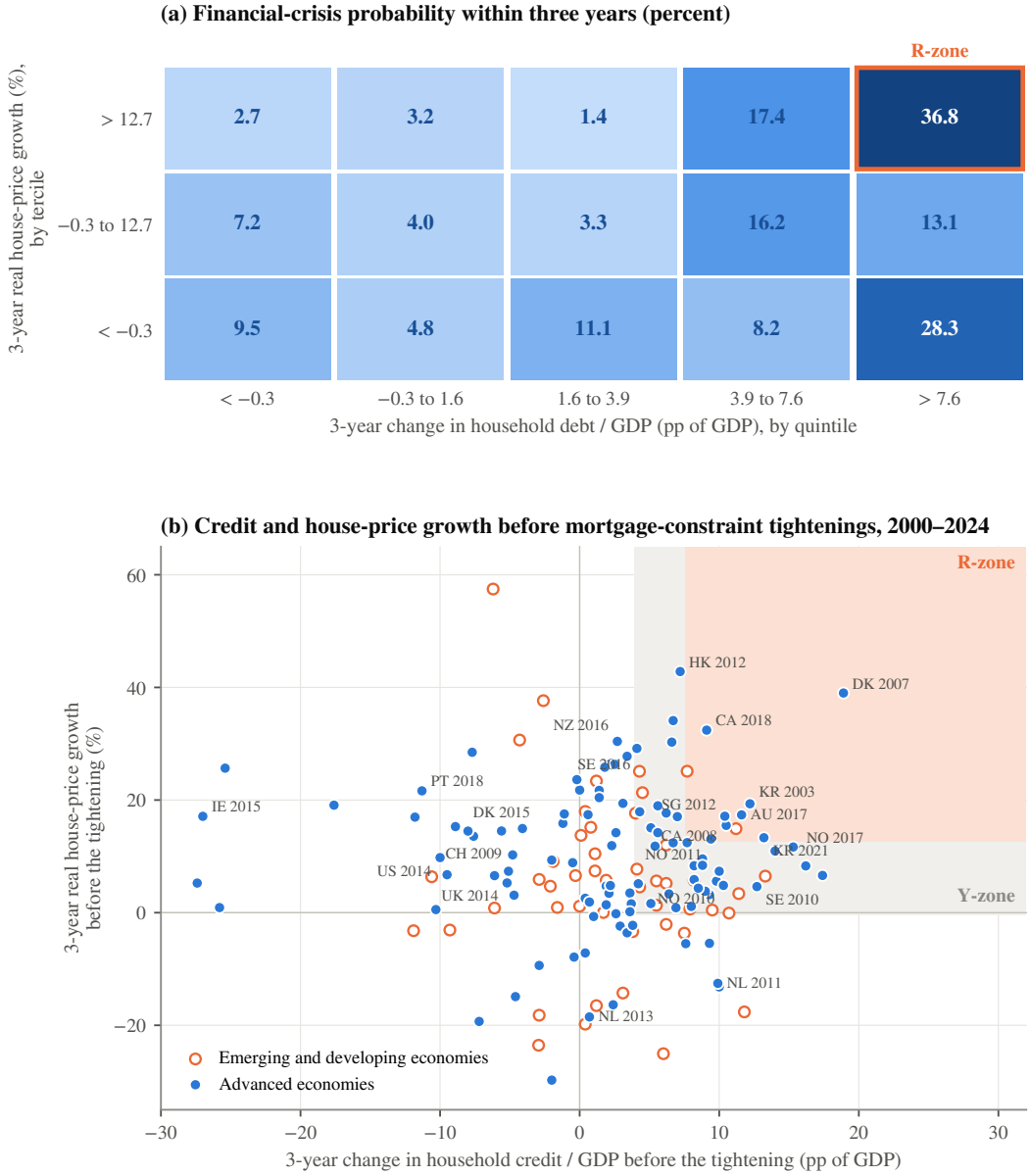


Figure 4: Crisis risk and mortgage-constraint tightenings on the credit and house-price map

Notes. Panel (a) reproduces Table 3, Panel D of **?**. Each cell reports the probability, in percent, that a financial crisis begins within three years for country-years sorted by the three-year change in household debt relative to GDP and three-year real house-price growth. The sample covers 42 countries from 1950 to 2016; crisis dates are from **?**. The outlined cell is the household R-zone. Cell values are conditional crisis probabilities and should not be interpreted as causal policy effects. Panel (b): each marker represents a tightening episode involving an LTV limit, an income-based limit, or a household loan restriction recorded in the IMF iMaPP database (**?**). Consecutive tightening actions within twelve months are treated as one episode. Positions are measured at the end of the quarter before the first action using the three-year change in credit to households relative to GDP and the three-year change in the BIS real residential property-price index. The shaded regions are the Y-zone and, nested within it, the R-zone, defined using the thresholds in **?**. Labels identify selected advanced-economy episodes, including the reforms studied in the household-level literature reviewed in Section **??**. Observations from 2020 to 2024 are affected by unusually large movements in nominal GDP.

growth is associated with potentially large corrective benefits. However, given the non-linearity in crisis risk discussed above, a relevant question for assessing the potential corrective benefits is whether the observed tightenings typically occur while countries are experiencing the combinations of credit and house-price growth associated with elevated historical risk.

Panel (b) of Figure **??** places the 154 tightening episodes recorded in the IMF iMaPP database between 2000 and 2024 and across 38 countries on a "crisis map" using the country's position in the quarter before the recorded action date. Tightening inside the R-zone is in fact uncommon: only 10 of the 154 episodes occur there, including 8 of the 105 episodes in advanced economies. House-price growth is in the top tercile in 56 episodes, but only 10 of these also have household credit growth in the top quintile, and household debt relative to GDP is falling in 46 episodes.

That most tightenings occur outside the R-zone does not necessarily imply that their corrective benefits are small. Mortgage constraints are preventive and forward-looking, and optimal implementation timing might very well be at intermediate credit and house-price growth levels. That few episodes fall inside the R-zone is thus consistent both with policies rarely being used when historical risk is actually at its highest *and* with successful preemption. Distinguishing the two requires a credible estimate of the path that credit, debt service, lending standards, and house prices would have followed without intervention, not only the economy's position at implementation.

As a final point, we note that while the crisis-prediction literature discussed above has focused mainly on credit, the micro literature has increasingly emphasized that financial vulnerability depends not only on the amount of debt, but also on the composition of household balance sheets. Households with substantial housing wealth may remain vulnerable if they hold little liquid wealth or face large required debt payments (**??**). Consistent with this, **?** find that improving borrowers' short-run liquidity has much larger effects on default and consumption than reducing mortgage principal without lowering current payments. Credit growth and leverage are therefore important indicators of crisis risk, but they do not fully capture households' financial vulnerability.

6.2 Changes in stability-relevant outcomes

The previous subsection highlighted the key role of credit as a crisis predictor, echoing the stated policy objectives. The household-level studies in Section **??** imply a modest decline in aggregate credit growth, while the macro studies reviewed in Section **??** find larger effects. Mortgage constraints also appear to lower house-price growth in markets where the rules bind strongly, although the evidence is less consistent at the national level. These changes move two central predictors of financial crises in the direction of lower risk.

Household-level evidence provides more detail on the composition of the credit response. Much of the reduction in mortgage debt occurs through the extensive margin, as some households delay or abandon a purchase and therefore do not originate a mortgage. Among households that

continue to buy, mortgage constraints generally reduce leverage and required debt payments. Both responses should make household balance sheets more resilient to declines in income or house prices. Some continuing buyers, however, meet larger down-payment requirements by drawing down liquid assets. This creates a leverage–liquidity trade-off. The household has less debt but may also have a smaller buffer available when income falls.

Evidence on consumption smoothing illustrates this trade-off. **?** find that Dutch homeowners affected by the LTV limit are better able to smooth consumption, as proxied by car ownership, once their liquidity buffers have been rebuilt. Affected Norwegian homeowners instead reduce debit-card spending more following job loss (**?**). The difference is consistent with the persistence of the liquidity response. Liquid assets recovered relatively quickly among Dutch buyers but remained lower for at least three years among Norwegian buyers.

The evidence on repayment outcomes points more consistently toward lower vulnerability, although the magnitude varies. **?** calculate that applying the US debt-to-income rule to the whole mortgage market would have lowered the five-year default rate of the 2007 and 2008 cohorts by 0.35 to 0.40 percentage points, small relative to default rates of 24 to 34 percent among loans not repaid early. **?** find that Dutch households affected by the LTV limit were 0.8 percentage points less likely to enter arrears, against a baseline rate of 3.3 percent. The effect was concentrated among below-median-income households. **?** examine resilience following an aggregate shock. After the Brexit referendum, low-income borrowers in areas more exposed to the UK loan-to-income limit defaulted roughly one-fifth to one-third less often than borrowers in less exposed areas.

Taken together, the evidence suggests that mortgage constraints move most observed components of Z toward lower financial vulnerability. Credit growth, mortgage originations, leverage, and default risk decline, while house-price growth may also slow in the most exposed markets. Some forces can work in the opposite direction, however. Liquid buffers may fall among households that continue to buy, and lenders may redirect credit toward risks outside the regulated segment.

6.3 Reduced-form effects

A small literature estimates the direct effect of mortgage constraints on the distribution of future GDP growth. This is commonly referred to as the "GDP-at-risk" approach. The evidence from the GDP-at-risk literature broadly indicates that macroprudential tightening reduces risks to future growth, but is less conclusive about which instruments provide this protection.

? find that tighter LTV limits narrow the distribution of future GDP growth by raising its lower tail and lowering its upper tail, without reducing mean growth.[9] **?** likewise estimate

[9]The mean effect is not zero everywhere, however. Using LTV changes in 56 economies, **?** estimate that a 10-percentage-point reduction in the maximum LTV lowers output by about 1.1 percent over four years, comparable to a 25-basis-point increase in the policy rate. The estimate is imprecise and confined to emerging economies. For advanced economies, the output cost of a tightening is indistinguishable from zero.

net benefits from macroprudential tightening and find borrower-based measures particularly beneficial when credit vulnerabilities are high. Using narratively identified shocks in twelve European economies, however, **?** find that the statistically significant improvement in the lower tail is driven mainly by lender-based measures. Taken together, the studies support a stabilizing role for macroprudential policy overall, but do not establish that borrower-based measures are consistently the main source of improved growth-at-risk.

6.4 What the evidence establishes about term (III)

The crisis literature provides the clearest evidence on the first component of term (III), that is, the impact of stability-relevant variables on financial stability $\nabla\Psi(Z)$. Rapid household credit growth is a robust predictor of financial distress, particularly when it coincides with rapid house-price growth, and credit-intensive booms are followed by deeper downturns. Other features of household balance sheets have received less attention in the crisis-prediction literature. The available evidence nevertheless suggests that the allocation of debt, required debt payments, lending standards, and household liquidity also matter for how strongly households respond when income or house prices fall.

Mortgage constraints appear to move most of these stability-relevant outcomes Z in a favorable direction. Aggregate credit growth falls, fewer mortgages are originated, and households that continue to buy generally take on less leverage. House-price growth also declines in some of the markets most exposed to the constraints, and the evidence on arrears and defaults points toward moderately improved resilience. Not every response reduces vulnerability, however. Some continuing buyers draw down liquid assets, and lenders may redirect credit toward other borrowers, regions, or asset classes.

Reduced-form evidence on the full term (III) from the GDP-at-risk literature points in the same direction. Macroprudential tightening tends to improve the lower tail of future GDP growth, although the specific contribution of mortgage constraints is less firmly established. Overall, existing research supports a positive corrective effect, but does not yet provide a quantitative estimate of term (III) for mortgage constraints. Such an estimate would require a causal mapping from each change in Z to external losses, as well as evidence on how that mapping varies with initial credit growth, house-price growth, and other financial conditions.

7 Directions for future research

The framework helps identify where additional research has the highest return. The best-measured object in the literature, the direct credit response of exposed households, is only a small part of the total response, and it does not reveal the full private burden of the policy, the equilibrium adjustments that follow, or the corrective benefit. In our view, one more estimate of this response is therefore less valuable than research on what happens to households

that are kept out of the market, on how much of the aggregate response comes from market adjustment, on where displaced credit goes, and on how large the impact on the external costs of household leverage actually is. We summarize this in the form of six areas for future research.

Follow households that do not obtain a mortgage. Much of the direct credit response comes from the extensive margin, yet most of what we know about adjustment comes from households that still buy. A key open question is therefore whether mortgage constraints delay homeownership by a year or two or exclude some households permanently. This distinction determines how the extensive margin should be interpreted in term (I). Linking mortgage applications and rejections to credit registers, housing transactions, rental contracts, and data on household and parental wealth would show how long entry is delayed, how much delayed households pay in rent, what and where they eventually buy, and whether family resources determine who enters (**??**). Such data exist, for instance, in the Nordic countries. Followed over a longer horizon, the same data could also show whether tighter constraints change the age profile of homeownership and the transmission of housing wealth across generations. Since households that never buy do not reveal their willingness to pay for borrowing capacity through their choices, survey designs that elicit it directly (**?**) are a natural complement.

Estimate direct and equilibrium effects in the same setting, and identify the channel. The aggregation gap in Section **??** is based on comparing studies of different countries, instruments, outcomes, and horizons, and on amplification estimates borrowed from other credit shocks. The tools needed to do better exist, but have rarely been applied to mortgage constraints. A study that estimates the household response within markets and the market-level response across markets with different average exposure, for the same outcome and horizon, delivers a regional multiplier directly (**??**). Such studies should also report the diagnostics for heterogeneous spillovers suggested by **?** and **?**, which, to our knowledge, no study of a borrower-based measure has done. Mortgage constraints also offer a natural test of the mechanism. The collateral channel operates under LTV limits but not under income-based limits, so house-price spillovers to credit should be weaker for LTI-constrained households. Settings where both types of limits were introduced, such as Ireland, can be used to test this. Loan-level data with lender identifiers would make it possible to separate propagation through housing markets from propagation through lenders (**?**), and estimates of the segmentation between owner-occupied and rental markets outside the United States (**?**) would help discipline the step from regional to national effects.

Trace where the displaced credit goes. A tightening only lowers financial vulnerability if the credit it removes does not reappear elsewhere in an equally fragile form. Section **??** discusses several examples of such leakage, including Irish banks expanding riskier corporate lending and mortgage lending in the United Kingdom (**??**). Households also find ways around the constraints, for instance by paying a large premium for unregulated debt (**?**) or through parents borrowing against their own homes (**?**). Most studies, however, measure lending in the

regulated segment only. The relevant outcome for term (III) is the distribution of leverage and debt service across the whole system, including debt held by parents, by consumer-credit and non-bank lenders, and by foreign affiliates. Credit registers covering all lenders and household balance-sheet data that link generations make this measurable in several countries. Without such evidence, a fall in regulated mortgage credit is only weak evidence of a fall in aggregate vulnerability.

Study households when downturns occur and measure external costs of high leverage in a downturn. The main rationale for mortgage constraints is that high household leverage imposes costs on others. Several authorities explicitly refer to an aggregate demand externality, where highly indebted households cut consumption sharply after a shock and thereby reduce demand and incomes for everyone else (Section **??**). Yet this externality is rarely measured directly. The evidence reviewed in Section **??** shows that credit growth predicts crises and deeper recessions, but this association does not separate the costs that indebted households bear themselves, which are already reflected in their borrowing decisions, from the costs they impose on others. Measuring the externality requires two steps. The first is to establish whether households whose mortgages were originated under tighter constraints actually cut consumption less when house prices fall or incomes decline. The general evidence that more indebted households cut spending more in downturns is strong (**??**), but the evidence for households affected by mortgage constraints is mixed, since lower leverage comes together with lower liquid buffers (**??**). Moreover, the aggregate downturns studied so far, the Brexit referendum and the Covid-19 pandemic, were mild for housing markets. The second step is to establish whether lower spending among indebted households reduces incomes and employment among other households. Regional evidence from the US housing bust and the Hungarian foreign-currency debt crisis suggests that such spillovers through local demand can be large (**??**), but we are not aware of studies that estimate them in a setting with mortgage constraints. The designs discussed above could be used for this purpose, by linking regulatory exposure at origination to outcomes in a later downturn for both exposed households and non-exposed households in the same local market. Since the size of the externality depends on whether monetary policy can offset the fall in demand (**??**), comparing countries inside and outside a currency union, or periods with and without a binding lower bound on interest rates, would also be informative.

Understand whether mortgage constraints keep countries out of the "R-zone." Few tightenings occur in the joint credit and house-price boom where historical crisis risk is highest. As discussed in Section **??**, this is consistent both with policy being used too late and with successful preemption. Separating the two requires an estimate of the path that credit, debt service, lending standards, and house prices would have followed without the tightening. The main obstacle is endogeneity, as authorities tighten in response to the very credit growth they cite as their objective. Policy series therefore need to be purged of their predictable component or identified narratively (**?**). With more than 150 tightening episodes across nearly 40 countries, event-study and synthetic-control designs on the crisis map are now feasible. The question such

designs should answer is whether a tightening keeps an economy from entering the region where the link between vulnerability and crisis risk in term (III) is steepest.

Evaluate mortgage constraints jointly. Most studies evaluate one instrument at a time, but many countries use several at once. Norway combines LTV and debt-to-income limits with amortization requirements and an interest-rate stress test, Ireland introduced LTV and LTI limits together in 2015, and Sweden combines amortization requirements with an LTV limit. As discussed in Section **??**, the instruments bind for different households, and the constraint that binds for a given household can change with house prices, income, and interest rates. The effect of tightening one instrument therefore depends on the others in place. A tighter LTV limit may, for instance, have little effect on a household that is already constrained by an income-based limit, but bind more often after house prices fall. The instruments also induce different adjustments, as LTV-constrained households draw on their savings while LTI-constrained households buy cheaper homes (**?**). Meta-analytic estimates suggest that the credit effect is larger when LTV and income-based limits are used together (**?**), but few studies show how the instruments interact for the same population of borrowers. Loan-level data that record all the relevant ratios at origination would show how many households are constrained by each limit, and by several limits at once, and how the response to a change in one limit depends on the others. Such evidence is needed to evaluate the overall stance of borrower-based policy rather than each instrument in isolation. A natural extension is also to bring bank capital requirements, such as risk-weight floors on mortgages or general capital requirements, into the same analysis, since they are often used alongside borrower-based limits and target the same vulnerabilities (**?**).

8 Conclusion

Mortgage constraints are usually introduced to restrain credit and household leverage, but their effects extend beyond the amount borrowed. Exposure depends on the policy instrument, household resources, and lender implementation. Households respond through debt, liquid savings, housing choices, and the timing of homeownership. The direct credit response is modest and comes in large part from fewer mortgage originations.

Aggregate estimates are larger than the household-level results can explain mechanically. Housing markets, lenders, and expectations provide plausible amplification channels, and amplification within the range found in related credit settings can account for much of the difference. These market responses also spread the incidence to existing owners, renters, lenders, and households that receive reallocated credit. The corrective benefit depends on whether the resulting changes prevent the build-up of vulnerable balance sheets. Historical crisis risk is highest during joint credit and house-price booms, while most tightenings occur outside that state.

A decline in credit therefore shows that the policy changed borrowing, but not whether it improved welfare. That assessment also requires the value of the borrowing capacity households

lose, the incidence of equilibrium changes, and the path financial vulnerability would have followed without the policy. The evidence reviewed here identifies many of these components and some areas for future research where the gain from more knowledge is high.

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A Data and coding behind Figure ??(b)

Panel (b) of Figure **??** is constructed from the 2017 Macroprudential Policies Evaluation Database (MaPPED) from **?**. MaPPED documents the life cycle of macroprudential and closely related policy instruments in all EU member states and the United Kingdom, recording for each policy action its instrument category and subcategory, the market segment targeted, the type of action (activation of a new tool, change in level, change in scope, deactivation), whether it tightened or loosened the stance, its main character (macroprudential, microprudential, or other), and a multi-select field, "Objective of the measure," in which the authority's stated motivation is recorded. The objectives field is populated from national regulatory documents by the country experts who compiled the database.

We keep policy actions whose main character includes "Macroprudential" and whose category is "Lending standards restrictions," which is MaPPED's label for borrower-based instruments, and then restrict to actions whose market segment includes residential real estate. This yields 113 actions in 18 countries. The sample comprises 49 loan-to-value limits, 28 debt-service-to-income limits, 12 maturity and amortization restrictions, 5 other income requirements for loan eligibility, 2 debt-to-income limits, 1 loan-to-income limit, and 16 other restrictions on lending standards. Of these, 89 are tightenings and 58 are activations of a new tool. Ninety-seven are classified as purely macroprudential, with the remainder carrying a joint macroprudential and microprudential (or other) character. Announcement years, where recorded, range from 1982 to 2017, with 71 of the 91 dated actions falling in 2010 or later. The unit of observation is the policy action rather than the country-instrument, so countries that adjusted a given limit repeatedly contribute several observations.

MaPPED's objectives field is multi-select and unranked. A measure may list several objectives, but the database does not identify the primary one. We therefore report the share of measures citing each objective. The mapping in Figure **??**(b) into four broad categories plus a residual is as follows. *Credit and household leverage* collects "General credit growth," "Sectoral credit growth," and "Household leverage." *House prices* corresponds to "Developments in asset prices." *Financial stability / resilience* collects "Other addressing the resilience of the financial system," "Currency risks," "Interconnectedness / contagion risks," "Market liquidity," and "Exposure concentration." *Bank sector* collects "Bank leverage," "Maturity mismatches on banks' balance sheets," and "Bank-level oversight (microprudential)." *Other* collects "Level playing field / competition," "Harmonisation," and "Other." We count a measure in a category if it cites any objective belonging to it, so categories overlap and shares do not sum to one.

We place "Currency risks" under financial stability rather than credit, on the grounds that the foreign-currency mortgage measures in Central and Eastern Europe that account for most of these cases were motivated by household exposure to exchange-rate shocks rather than by the volume of borrowing as such. Moreover, we place "Bank-level oversight (microprudential)" under the bank sector. Neither affects the ranking in Figure **??**(b).

B Derivation of the organizing equation

This appendix derives equation (**??**) and states the assumptions behind it. The framework starts with optimizing households and adds a reduced-form external cost of financial vulnerability of the kind studied in the normative macroprudential literature (**?????**). The purpose is not to calculate welfare, but to show how the objects discussed in the main text enter a first-order welfare expression.

The notation and sign convention are the same as in the main text. A higher τ permits more borrowing, so a tightening has $\mathrm{d}\tau < 0$. Statutory capacity is κ_i, effective capacity is $\tilde{\kappa}_i$, and implementation pass-through ϵ_i is defined in equation (**??**). We use η_i for the Lagrange multiplier on the borrowing constraint in utility units and λ_i for the same object in money units.

B.1 Environment

Households and markets. There is a unit continuum of households indexed by i. Household i chooses tenure (own or rent), a consumption plan, housing h_i (quantity, quality, and location), mortgage debt b_i ($b_i = 0$ for renters), liquid saving, and other financing such as unsecured credit or family transfers. We collect these choices in the vector a_i, and preferences are $U_i(a_i; Z)$, where Z is defined below.

Households take as given a vector of market outcomes $x = (x_1, \ldots, x_J)$, which includes the house price P, the rent R, mortgage spreads and fees s, and lender underwriting standards beyond the statutory rule. The budget constraint is written as $B_i(a_i; x) \leq 0$, where B_i is the present value of expenditure less the present value of resources. For a buyer, B_i includes the purchase price Ph_i, less the mortgage b_i, plus the present value of debt service. For a renter, it includes the present value of rent. Payments at later dates are valued at the household's own discount factor, so all objects below are measured in units of the numéraire at the date of the policy change.

B.1.1 Borrowing constraints

Household i faces the constraint

$$b_i \leq \tilde{\kappa}_i(\tau, x), \tag{B.1}$$

where effective capacity $\tilde{\kappa}_i$ is statutory capacity κ_i adjusted for how the rule is implemented. The instruments in the main text differ in which variables enter κ_i:

- *LTV limit:* $\kappa_i = \tau P h_i$, where τ is the maximum loan-to-value ratio. Capacity rises with the house price, $\partial\kappa_i/\partial P = \tau h_i > 0$, which is the collateral constraint of **?**.
- *LTI limit:* $\kappa_i = \tau y_i$, where y_i is the income the lender uses. Capacity does not depend on the house price. A DTI limit has the same form applied to total debt, so mortgage capacity is τy_i less the household's other debt.

- *DSTI limit:* required payments may not exceed a share τ of income. With payments equal to $\alpha(r+s,T)\,b_i$, where α is the annuity factor for the mortgage rate $r+s$ and maturity T, capacity is $\kappa_i = \tau y_i/\alpha(r+s,T)$. Because α increases with the rate, $\partial\kappa_i/\partial r < 0$ and $\partial\kappa_i/\partial s < 0$.
- *Amortization requirement:* the requirement restricts the repayment path rather than the loan at origination. It gives one constraint per period, and λ_i below is the present value of the money-metric multipliers on these constraints. A household can therefore be constrained by an amortization requirement while having unused capacity at origination, as in the Swedish evidence (**?**).

Implementation enters through effective capacity. With lender exemptions e_i, for instance allocated under a quota (**?**), effective capacity is $\tilde{\kappa}_i = \kappa_i + e_i$. If lenders grant more exemptions when the rule is tightened, $\partial e_i/\partial\tau < 0$, and equation (**??**) gives

$$\epsilon_i = 1 + \frac{\partial e_i/\partial\tau}{\partial\kappa_i/\partial\tau} < 1.$$

Lender underwriting standards enter $\tilde{\kappa}_i$ in the same way, which is why we include them in x.

B.1.2 The household problem

Conditional on owning, household i solves

$$V_i^o(\tau,x,Z) = \max_a\ U_i(a;Z) \quad \text{s.t.} \quad B_i(a;x) \le 0\ \ [\mu_i], \qquad b(a) \le \tilde{\kappa}_i(\tau,x)\ \ [\eta_i], \tag{B.2}$$

where $\mu_i > 0$ is the marginal utility of the numéraire and $\eta_i \ge 0$ is the utility value of one more unit of borrowing capacity. The value of renting, $V_i^\rho(x,Z)$, is defined in the same way without the mortgage. The household's value is $V_i = \max\{V_i^o, V_i^\rho\}$, and $S_i \equiv V_i^o - V_i^\rho$ is its ownership surplus.

B.1.3 Aggregates and the external cost

The vector Z collects aggregate outcomes associated with financial vulnerability, such as total household credit $\int b_i\,\mathrm{d}i$, the distribution of leverage and debt service, the composition of new lending, and house-price growth. These outcomes affect households, for instance through the probability and depth of a financial crisis, but each household takes them as given. The external cost $\Psi(Z)$ is the money-metric value of these effects summed over households. Only its gradient enters the derivation, so we define it directly:

$$\nabla\Psi(Z) \equiv -\int \frac{1}{\mu_i}\,\frac{\partial V_i}{\partial Z}\,\mathrm{d}i. \tag{B.3}$$

As an example, suppose a crisis occurs with probability $\pi(Z)$ and that

$$U_i(a; Z) = \big(1 - \pi(Z)\big)\, U_i^{NC}(a) + \pi(Z)\, U_i^{C}(a),$$

where U_i^{NC} and U_i^{C} are utility in the no-crisis and crisis states. Then $\partial V_i/\partial Z = -\nabla\pi\,(U_i^{NC} - U_i^{C})$ at the household's optimum, and

$$\nabla\Psi(Z) = \nabla\pi(Z)\,\mathcal{C}, \qquad \mathcal{C} \equiv \int \frac{U_i^{NC} - U_i^{C}}{\mu_i}\,\mathrm{d}i \;\geq 0.$$

The external cost thus equals the change in crisis probability times the money-metric loss in a crisis, summed over households. Households take their own losses in a crisis into account when they choose how much to borrow, given π. What they do not take into account is the effect of their borrowing on π. The evidence that credit-financed booms are followed by deeper recessions (**??**) bears on $\mathcal{C}$, and the evidence that credit growth predicts crises (**??**) bears on $\nabla\pi$. Nothing requires a binary crisis state, and if crisis severity also depends on Z, equation (**??**) includes that effect.

The market outcomes x and the aggregates Z are determined in equilibrium by housing and rental market clearing, lender behavior, and expectations. We do not solve for this equilibrium. We only require that the equilibrium path $x(\tau)$, $Z(\tau)$ is differentiable, and we treat $\dot{x} \equiv \mathrm{d}x/\mathrm{d}\tau$ and $\dot{Z} \equiv \mathrm{d}Z/\mathrm{d}\tau$ as the responses estimated in the empirical literature. These are present values of the response paths, and λ_i is the present value of the multipliers on the borrowing constraint at each date and in each state.

B.1.4 Assumptions

Assumption B.1 (Price taking)**.** *Households and lenders take x and Z as given.*

Assumption B.2 (Envelope theorem)**.** *Conditional on tenure, the envelope theorem applies to problem* (**??**) *(**?**). At regulatory thresholds the derivatives are one-sided, and we use the derivative in the direction of a tightening. Households bunching at a threshold are constrained households with $\eta_i > 0$.*

Assumption B.3 (Rational borrowing)**.** *Households maximize their own welfare with correct beliefs. We relax this in Section* **??**.

Assumption B.4 (Marginal reform)**.** *The tightening $\mathrm{d}\tau < 0$ is small and permanent.*

Assumption **??** is the source of the policy rationale. Because no household or lender takes into account the effect of its own borrowing on Z, private optimization does not lead to the social optimum, and the external cost remains in the welfare expression. We abstract from compliance and screening costs and from lender profits other than those captured by spreads.

B.2 Derivation

Step 1: Envelope conditions. Fix tenure at the household's optimum. The Lagrangian of problem (**??**) is

$$\mathcal{L}_i = U_i(a; Z) - \mu_i\, B_i(a; x) + \eta_i\big[\tilde{\kappa}_i(\tau, x) - b(a)\big].$$

By Assumption **??**, the derivatives of the value function with respect to τ, x, and Z equal the partial derivatives of $\mathcal{L}_i$ at the optimum. Changes in the household's own choices have no first-order effect on its value:

$$\frac{\partial V_i}{\partial \tau} = \eta_i \frac{\partial \tilde{\kappa}_i}{\partial \tau} = \eta_i\, \epsilon_i \frac{\partial \kappa_i}{\partial \tau}, \tag{B.4}$$

$$\frac{\partial V_i}{\partial x_j} = -\mu_i \frac{\partial B_i}{\partial x_j} + \eta_i \frac{\partial \tilde{\kappa}_i}{\partial x_j} = \mu_i\, m_{ij} + \eta_i \frac{\partial \tilde{\kappa}_i}{\partial x_j}, \tag{B.5}$$

where the second equality in (**??**) uses equation (**??**), and

$$m_{ij} \equiv -\frac{\partial B_i}{\partial x_j} \tag{B.6}$$

is household i's net position in market j. For the house price, m_{ij} is net housing sold, negative for buyers and positive for sellers. For the rent, it is net rental supply, negative for renters and positive for landlords. For the mortgage spread, it is negative and proportional to b_i for borrowers and positive for the owners of lenders. The derivative with respect to Z is $\partial V_i/\partial Z = \partial U_i/\partial Z$, which enters equation (**??**).

Step 2: Money metric. We measure welfare in units of the numéraire by weighting each household by $\omega_i \equiv 1/\mu_i$, evaluated before the policy change, so that one unit of the numéraire has the same value whoever receives it. Welfare is

$$W(\tau) = \int \omega_i\, V_i\big(\tau, x(\tau), Z(\tau)\big)\, \mathrm{d}i, \tag{B.7}$$

and the money-metric shadow value of borrowing capacity is

$$\lambda_i \equiv \frac{\eta_i}{\mu_i} \geq 0. \tag{B.8}$$

Step 3: Aggregate. Differentiating equation (**??**) along the equilibrium path and substituting equations (**??**)–(**??**) gives the organizing equation.

Proposition B.1 (Organizing equation)**.** *Under Assumptions* **??**–**??**,

$$\frac{\mathrm{d}W}{\mathrm{d}\tau} = \underbrace{\int \lambda_i\, \epsilon_i \frac{\partial \kappa_i}{\partial \tau}\, \mathrm{d}i}_{\text{(I) Direct effect}} + \underbrace{\sum_j \left[\int \Big(m_{ij} + \lambda_i \frac{\partial \tilde{\kappa}_i}{\partial x_j}\Big)\mathrm{d}i\right] \dot{x}_j}_{\text{(II) Equilibrium incidence}} - \underbrace{\nabla\Psi(Z)\cdot\dot{Z}}_{\text{(III) Corrective effect}}. \tag{B.9}$$

Proof. By the chain rule, $\mathrm{d}W/\mathrm{d}\tau = \int \omega_i\big[\partial V_i/\partial \tau + \sum_j (\partial V_i/\partial x_j)\,\dot{x}_j + (\partial V_i/\partial Z)\cdot\dot{Z}\big]\mathrm{d}i$. Substi-

tuting equations (**??**) and (**??**), using $\omega_i \mu_i = 1$ and $\omega_i \eta_i = \lambda_i$, gives terms (I) and (II). The last term equals $-\nabla \Psi \cdot \dot{Z}$ by equation (**??**). □

Multiplying equation (**??**) by $\mathrm{d}\tau$ and using $\mathrm{d}x_j = \dot{x}_j \, \mathrm{d}\tau$ and $\mathrm{d}Z = \dot{Z} \, \mathrm{d}\tau$ gives equation (**??**) in the main text. Grouping the same terms by whether a household is constrained ($\lambda_i > 0$) or not, instead of by term, gives a decomposition into an effect on constrained households, an effect on unconstrained households, and crisis prevention. The two decompositions add up the same terms in a different order. Ours follows the objects that the different empirical literatures estimate.

B.3 Properties used in the main text

Household adjustments. No behavioral response appears in equation (**??**). When a household responds to a tightening by drawing down liquidity, buying a cheaper home, delaying a purchase, or relying on family support, the change in its welfare is fully captured by the shadow value λ_i times the change in capacity, because the household was already optimizing. The adjustments documented in Section **??** should therefore not be added to term (I) as separate costs. They remain informative, however, as they show that the constraint binds, they help discipline estimates of λ_i, they change the composition of debt that enters Z, and they generate the market responses $\dot{x}$.

Transfers and constraint revaluation in term (II). Each market response enters term (II) through two channels. The first, $\int m_{ij} \, \mathrm{d}i$, is a transfer between households. Let n_{ijt} be household i's net sales in market j at date t and Q_{it} its discount factor for date t, with $Q_{i0} = 1$, so that $m_{ij} = \sum_t Q_{it} \, n_{ijt}$. Market clearing requires $\int n_{ijt} \, \mathrm{d}i = 0$ at each date, so that

$$\int m_{ij} \, \mathrm{d}i = \sum_{t \geq 1} \mathrm{Cov}\left(Q_{it}, \, n_{ijt}\right). \tag{B.10}$$

For a price paid once at the date of the policy change, such as the purchase price of a home, the transfer channel sums to zero and affects only who gains and who loses. For payments spread over time, such as rents and mortgage spreads, the sum is zero only if households on the two sides of the market value future payments equally. Constrained households value future payments less: in a two-period model in which household i can borrow at rate r, $Q_{i1} = (1 - \lambda_i)/(1 + r)$. The covariance in equation (**??**) is the distributive pecuniary externality of **?**, and it is small when the exposed households on the two sides of a market have similar shadow values. The main text treats the first channel as a transfer, which is exact for house prices and an approximation for rents and spreads.

The second channel, $\int \lambda_i \, \partial \tilde{\kappa}_i / \partial x_j \, \mathrm{d}i$, does not net out. Under an LTV limit, $\partial \tilde{\kappa}_i / \partial P = \tau h_i$, so a lower house price reduces capacity for every constrained household. The two channels can work in opposite directions for the same household. For a first-time buyer with $m_{iP} = -h_i$, the

net exposure to the house price is $-h_i(1-\lambda_i\tau) < 0$, so a lower price is still a gain, although a smaller one than the transfer channel alone suggests. For an owner who sells a home of size h_i^0 to buy a home of size h_i, the net exposure is $-(h_i - h_i^0) + \lambda_i\tau h_i$, which is positive when the household trades up only modestly and is constrained, so a lower price is a loss (**?**). Under LTI and DTI limits, $\partial\tilde{\kappa}_i/\partial P = 0$ and only the transfer channel remains.

Households displaced from ownership. Under free tenure choice and a smooth distribution of ownership surplus, a small tightening moves only households that are close to indifferent between owning and renting ($S_i \approx 0$). Their welfare loss is second order, because the envelope theorem applies to the chosen tenure, while their effect on credit is first order, because each of them removes an entire loan. If instead households with $S_i > 0$ are kept out of ownership by a down-payment requirement, a rejected application, or an exhausted exemption quota, their loss is first order and is not captured by λ_i. Let $n_R(\tau)$ be the mass of such households and $\bar{s}_R$ the average money-metric surplus S_i/μ_i of those who are newly kept out by the tightening. The tightening then adds

$$-\,\bar{s}_R\,\frac{\mathrm{d}n_R}{\mathrm{d}\tau}\,\mathrm{d}\tau < 0 \tag{B.11}$$

to the direct effect. Missing mass in bunching designs and entry responses, such as those in **?**, measure $\mathrm{d}n_R$ only if displaced households are rationed rather than close to indifferent. They are also silent on $\bar{s}_R$.

B.3.1 Internalities

Suppose households overvalue borrowing, for instance because they expect house prices to grow faster than they will or neglect repayment risk. Let households choose as if they maximized $\hat{U}_i = U_i + \mu_i\delta_i b_i$, where δ_i is the money-metric overvaluation of a marginal unit of debt, while welfare is evaluated with U_i. The envelope theorem applies to the problem households actually solve, so terms (I)–(III) are unchanged, with λ_i and m_{ij} evaluated at the households' choices. Since $U_i = \hat{U}_i - \mu_i\delta_i b_i$, welfare picks up one additional term:

$$\frac{\mathrm{d}W}{\mathrm{d}\tau} = \big[\text{right-hand side of (??)}\big] \qquad \underbrace{-\int \delta_i\,\frac{\mathrm{d}b_i}{\mathrm{d}\tau}\,\mathrm{d}i}_{\text{(IV) Internality correction}}\quad, \tag{B.12}$$

where $\mathrm{d}b_i/\mathrm{d}\tau$ is the total change in household i's borrowing, including the response to market outcomes. A tightening reduces borrowing, so with $\delta_i > 0$ term (IV) times $\mathrm{d}\tau$ equals $-\int \delta_i\,\mathrm{d}b_i\,\mathrm{d}i > 0$, a benefit. The behavioral response that drops out of the welfare expression under Assumption **??** now matters, as in the theory of corrective taxation. If overvaluation is concentrated among constrained borrowers, terms (I) and (IV) partly offset for the same households, and the net effect on these households can be positive if δ_i is large. Data linking house-price expectations to leverage choices (**?**) offer one way to measure δ_i.

B.4 Limits

The organizing equation has four limits. First, it is a first-order expression. Observed changes in LTV and income limits are often several percentage points, and the second-order terms are then not negligible. Larger reforms require a structural model. Second, the envelope argument does not hold when households are rationed out of ownership or borrow more than is good for them. We state both cases as separate terms, equations (**??**) and (**??**), but for down-payment constraints rationing may be common. Third, the equation adds money-metric gains and losses without distributional weights, and the transfer channel nets out exactly only for one-time prices. The main text therefore reports incidence by group, so that other weights can be applied. Fourth, the present-value convention compresses timing. Announcement effects, transition paths, and changes in beliefs are part of $\dot{x}$ and $\dot{Z}$ but are not modeled separately.

C A practitioner's guide to estimating general equilibrium amplification

This appendix first shows how a researcher with household-level estimates can calculate the implied effect on aggregate credit growth, which makes it possible to compare the micro- and macro-level effects. The second part provides an overview of methods that can be used to measure equilibrium amplification in order to get closer to an aggregate effect. **?** provide a similar guide.

C.1 Calculating the implied micro effect

The object that we are after is the reduction in annual aggregate credit growth, in percentage points, implied by the estimated household-level responses. The procedure is based on an accounting identity: apply the treatment effects to the affected segment of new originations and normalize by the outstanding debt stock, so that

$$\Delta(\text{credit growth}) \approx \Delta(\text{originations})/(\text{debt stock}).$$

Table **??** applies this mapping to **?**, **?**, and **?**. The calculation requires only four inputs beyond the estimates themselves.

The first input is incidence: the share θ of pre-reform originations, measured by volume, that would have violated the new limit. Bunching designs recover this input from the pre-reform distribution of LTV or DTI ratios, and predicted-treatment designs report it as the treated share. In the studies above it ranges from a small fraction of the jumbo segment in the United States to 59 percent of new mortgages in Norway. The second is the extensive-margin response e: the share of treated households whose purchase is eliminated, identified by the missing mass in a bunching design or by a difference-in-differences estimate on the purchase probability. The third is the intensive-margin response ι: the average proportional reduction in loan size among treated households that still purchase, inclusive of those who do not adjust. When a study reports that a share s of treated borrowers adjusts by an average conditional amount Δ, the relevant input is $\iota = s\Delta$. The fourth is a pair of aggregates from national statistics: the annual origination volume O of the regulated market segment and the outstanding stock B of household mortgage debt. The implied effect on annual credit growth, in percentage points, is then

$$\Delta g \;\approx\; -\theta \quad \overbrace{\left[e + (1-e)\,\iota\right]}^{\text{Extensive + intensive margin}} \quad \frac{O}{B} \times 100. \tag{C.1}$$

The bracket is the fraction of treated origination volume the policy removes, split into loans that disappear entirely (the extensive margin) and loans that shrink (the intensive margin). This reading requires e and ι to be volume-weighted, or baseline loan size to be unrelated to which households exit or adjust. Where studies report household-weighted responses, the bracket is an approximation. Multiplying by θ converts it into a fraction of all originations, and the ratio

Table C.1: Back-of-the-envelope aggregation of household-level credit responses

	?	**?**	**?**
Setting	United States; 2014 ATR/QM rule; high-DTI jumbo mortgages, extrapolated to the full purchase market	Norway; LTV caps of 90% (2010) and 85% (2012)	Finland; 95% LTV limit for first-time buyers, July 2016
Treated share θ	High-DTI share of jumbo originations	59% of new mortgages above the cap before the reform	≈50% of first-time buyers (pre-reform median LTV 96%)
Extensive margin e	15% of high-DTI jumbo borrowing eliminated (2% of the jumbo market)	6% of treated households do not purchase	Net 1,030 fewer first-time apartment buyers with a mortgage (missing minus excess mass)
Intensive margin $\iota = s\Delta$	20% of high-DTI loans shifted below the threshold, each by an illustrative 4.4% ($\iota \approx 0.009$; 2.7% of the jumbo market)	11% lower debt among treated buyers	€6,493 lower mortgage per treated buyer
Forgone originations	\$12.0bn (extensive) + \$0.7bn (intensive)	Not reported in currency; see note (b)	≈€150m (extensive, at €145,323 per buyer) + ≈€38m (intensive, over the ≈50% of 11,703 first-time buyers treated)
Debt stock B	\$8.17tn, US 1–4 family mortgage debt, 2014	Norwegian household debt (authors' calculation)	€86.8bn, household housing loans, 2017
Implied Δg (pp)	−0.15	−0.16	−0.22

Notes. Δg is the implied reduction in annual aggregate credit growth, computed as forgone originations divided by the outstanding debt stock (equation (**??**)), and reported as a one-year flow effect. (a) United States: extensive and intensive shares from Table 3 of **?**; debt stock from Federal Reserve Board, *Mortgage Debt Outstanding*, December 2014. The \$12.0bn extensive figure is the paper's extrapolation to the full \$600bn purchase market, as if the GSE exemption had lapsed. The \$0.7bn intensive figure is our calculation on the same basis, $0.027 \times 0.044 \times \$600\text{bn}$, using the paper's illustrative 4.4 percent loan reduction for the average borrower above the threshold. The realized effect on the jumbo segment alone is about \$0.6bn, or 0.01 percentage points. The combined extrapolated effect is $(12 + 0.7)/8{,}170 = 0.15$ percentage points. (b) Norway: the authors' own back-of-the-envelope calculation (Appendix C, equations 12–13 of the cited version) gives −0.27 percentage points but applies the difference-in-differences coefficients to the population-wide purchase probability, which implicitly sets $\theta = 1$. Rescaling by the reported pre-reform incidence of 59 percent gives −0.16. (c) Finland: extensive margin from the 2017 bunching estimates (Table 3), intensive margin from the difference-in-differences estimates conditional on buying (Table 7), and the treated share from the pre-reform median LTV (Table 2), all in the VATT Working Paper 148 version; debt stock from Statistics Finland, *Statistics on Indebtedness 2017*. The implied effect ranges from −0.17 to −0.23 percentage points across assumed treated shares.

O/B converts forgone originations into forgone credit growth. Table **??** reports the inputs and the implied effects for the three studies aggregated in the main text, following equation (**??**). In the US and Finnish cases the extensive margin supplies most of the total, but in Norway the two margins contribute similar amounts.

Researchers interested in this number should be aware of four potential errors in this type of calculation. The first error is applying treatment effects to the whole population rather than to the treated. The back-of-the-envelope calculation in Appendix C of **?** applies the difference-in-differences coefficients to the population-wide purchase probability, which implicitly sets $\theta = 1$ and yields −0.27 percentage points. Rescaling by the paper's own 59 percent incidence gives

-0.16. The second is mixing count shares with volume shares. Treated loans are systematically larger than average loans, since the treated are selected on high leverage or high debt relative to income, so a count-based θ understates the volume at stake unless it is converted using the average treated loan size. The third is a denominator mismatch. The stock B must match the credit aggregate used in the macro comparison, as household mortgage debt, total household credit, and total bank credit are different objects. **?**, for instance, standardize on annual bank credit growth, so a comparison against their benchmark requires the corresponding stock. The fourth is the horizon. Equation (**??**) delivers the effect on one year's flow of originations. If the researcher is interested in longer-term effects, they need to take a stand on how long the constraint continues to bind and on the persistence of the extensive-margin response.

The resulting number is a partial-equilibrium accounting object that excludes everything the market-time fixed effects absorb by construction, a problem commonly known as the missing intercept (**?**). Hence, the number is not the aggregate effect on credit, and it is not a policy counterfactual. We still encourage researchers to report this number, as it generally requires little extra data and would help with the comparison to other studies. Reporting the numbers from equation (**??**), a decomposition into margins, and a sensitivity range over the assumed treated share gives the literature a number that can be placed against other micro studies and against the macro benchmarks.

C.2 Estimating general equilibrium amplification

To provide a full estimate of the total effect of a policy, researchers need to estimate the total effect, including any general equilibrium amplification usually removed by fixed effects. This section provides an overview of recent methods that try to estimate the total effect of a policy in different ways.

Before we discuss how recent studies have attempted to estimate general equilibrium amplification or spillovers, we need a precise definition of what we are after. The general equilibrium multiplier is the ratio of the total effect to the direct effect, for a stated outcome, at a stated level of aggregation, over a stated horizon. The evidence in Section **??** shows that this multiplier differs by outcome, that it differs by the state of the market, and that it is endogenous to the policy itself. For example, **?** estimate that equilibrium effects accounted for 94 percent of the cross-sectional credit response to borrowers' distance from the Irish limits during the pre-introduction boom, but only 23 percent of the response to the limits themselves. This gives ratios of total to direct effects of roughly 17 and 1.3. The first number describes the credit and house-price feedback of the boom and the second the policy response, which is imprecisely estimated, and the authors caution that the decline cannot be attributed to the limits alone. In addition, the regional multiplier is different from the national multiplier, as discussed in Section **??**, and will also differ depending on the monetary policy response (**?**). A claim about "the" multiplier should therefore always say the multiplier of what, where, and when.

Direct estimation of the multiplier. The most immediately feasible approach requires no data beyond what existing micro studies already use. Suppose the outcome of household i in market m depends on its own exposure to the constraint and on an equilibrium object that is a function of average exposure in the market. When spillovers are homogeneous and linear, and own and market-average exposure are unrelated to household-specific trends and to other local shocks, a single regression of the outcome on own exposure and on the leave-out mean of exposure among other units in the same market, with time fixed effects, recovers both objects at once (**?**):

$$y_{imt} = \alpha_i + \delta_t + \beta\,(E_i \times \text{Post}_t) + \gamma\,(\bar{E}_{-i,m} \times \text{Post}_t) + \varepsilon_{imt}, \tag{C.2}$$

where E_i is the household's pre-policy exposure and $\bar{E}_{-i,m}$ is the mean exposure of other households in market m. The direct effect is β, the spillover is γ, and the ratio of total to direct effects is $(\beta + \gamma)/\beta$. Because δ_t absorbs effects common to all markets, this ratio is a regional rather than an aggregate multiplier. **?**, studying the introduction of interest-only mortgages in Denmark, estimate within-municipality spillovers from payment-constrained to leverage-constrained households through house prices using a design similar to this.

The equivalent two-regression version estimates the within-market effect under market-time fixed effects and then re-estimates on data collapsed to the market level with time fixed effects only, recovering the multiplier as the ratio (**?**). **?** implement this comparison for the Irish limits. The two procedures are closely related, although they coincide only under equal market weighting and homogeneous effects. The one-regression version of **?** delivers a standard error on the spillover directly, whereas inference on the multiplier itself requires the joint uncertainty of the total and direct estimates.

In practice the design requires three construction choices. The market definition should correspond to where the equilibrium object lives, which for housing spillovers means a local housing market such as a county, municipality, or commuting zone. The exposure measure should be defined before the policy, from the pre-reform distance to the limit or a prediction of violation based on pre-reform characteristics. Finally, the specification should be estimated on the same outcome, instrument, and horizon as the partial-equilibrium estimate to allow for the comparison.

Diagnostics. **?** shows that the leave-out-mean regression is biased when there are omitted spillover types, nonlinear direct effects, or measurement error. The decomposition is valid when spillovers are homogeneous and thus affect treated and untreated units alike. In that case, market-time fixed effects absorb the spillovers for the individual unit, and removing the market-time fixed effect estimates the effect with the spillovers included. However, this is a hard assumption to fulfill, especially for mortgage constraints. **?** show that homogeneity fails when the treatment acts as a binding constraint. Imagine, for example, a policy that loosens the LTV constraint, which creates spillovers to the untreated units through increases in house prices. Treated households at the LTV constraint will not be able to respond to rising house prices in the same way as the untreated households, as they are bound by the collateral constraint. More

generally, a household at its borrowing limit cannot respond fully to a market-level spillover, which the unconstrained neighbors can. In this case, **?** show that fixed effects can amplify rather than remove the bias, although this is not always the case. **?** recommend estimating the outcome as a function of the share of exposed households in the market separately for exposed and unexposed households, without market fixed effects, and plotting the two fitted lines together with the market average.

? adds a complementary test: the estimated spillover should be smaller for units less exposed to a hypothesized channel. A spillover that is equally large for less exposed units is likely to be mechanical. Under an LTI limit, for example, the collateral channel is absent, so the house-price spillover on credit should be weaker for LTI-constrained than for LTV-constrained households. Huber shows that omitted spillover types, nonlinear direct effects, and measurement error in exposure can all produce such mechanical bias, and that additional leave-out means for other channels, flexible functional forms, and an instrument for own exposure that is uncorrelated with the exposure of others can address it. The direction of the resulting bias in the multiplier depends on the channel. If the spillover runs through prices and lower prices reduce borrowing among unconstrained households, the comparison group moves with the treated, the within-market gap understates the direct effect, and the multiplier is overstated; the sign reverses if lower prices instead draw new buyers into the market. If lenders reallocate credit toward the comparison group, as the Irish evidence documents, the gap overstates the direct effect and the multiplier is understated. We are not aware of a study that reports these heterogeneity tests for a borrower-based measure.

Interpreting the output. The interpretation of the results depends on several factors that researchers ought to be aware of. First, when several spillover channels operate at once, a likely case when it comes to mortgage constraints, a single leave-out mean does not isolate each type of spillover, and cross-market channels contaminate the comparison across markets as well as within them. For example, mortgage constraints can plausibly generate spillovers through local housing demand, lender balance sheets (**?**), and the rental market (**?**).

Second, idiosyncratic, within-market measurement error in exposure attenuates the within-market coefficient more than the market-level one, because aggregation averages it out, and therefore inflates the ratio of the total to direct effects. Error in the market-level component of exposure biases the ratio in the opposite direction (**?**). Predicted-treatment designs guarantee some mis-measurement, since households respond with family transfers, cheaper properties, or different contract terms (**?**).

Third, the output is a regional multiplier, not the missing intercept. National equilibrium responses, including the policy rate, system-wide credit reallocation, and the regulator's reaction, sit in the time fixed effects. **?** show that regional amplification and aggregate dampening can coexist. A researcher interested in the national multiplier needs to turn to alternative approaches, with their own caveats and difficulties. We turn to these next.

Alternative approaches. Two alternatives do not rely on the within-market comparison and therefore avoid the contamination just described, although each carries assumptions of its own. The aggregate component can be recovered from regional data directly, either by combining a regional VAR with factor analysis (**?**) or by estimating local and aggregate effects jointly in a Bayesian framework that treats micro estimates as priors (**?**). Where mortgage lending is concentrated among a few lenders, as it is in most countries that have adopted these policies, the idiosyncratic shocks of large lenders can be aggregated into an instrument for aggregate outcomes (**?**). This is one of the few designs available when the policy variation is purely national and there is no cross-section to exploit, although it identifies propagation from a lender shock rather than from the policy itself.

Another approach is to try to estimate a single structural parameter that governs the spillover. For housing, **?** identify the degree of market segmentation as an important parameter that governs how credit affects house prices and thus generates spillovers. The degree of market segmentation is identified by the ratio of the price-rent response to the homeownership response to an identified credit shock (**?**). It has been estimated for the United States alone, and estimating it in the countries that supply the micro evidence would discipline the micro-to-macro extrapolation more than any additional household-level estimate.

The source of the amplification. For a researcher interested in identifying the source or mechanism behind the spillover, the mediation approach of **?** addresses that directly. The approach exploits the fact that the housing channel propagates within geographically segmented markets while the intermediary channel propagates over lender networks that span them. Estimating the source requires loan-level data with lender identifiers, which exist in several of the countries that supply the micro evidence, but it has not yet been applied to a policy shock.